

Transcendence and Undecidability in Hypergraph Turán Densities

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Abstract

Turán problems are central to extremal combinatorics: they ask how dense a large uniform hypergraph can be while avoiding prescribed configurations, with the limiting optimum called the Turán density. We prove that, for every sufficiently large uniformity r , some finite family of r -graphs has transcendental Turán density. For each fixed sufficiently large r , we also show that it is undecidable whether the Turán density of a given finite family of r -graphs is at most a given rational number, even under the promise that the density is at least that number. The first result answers a question of Jacob Fox, while the second answers a question of Oleg Pikhurko in the negative. For fixed uniformity, the undecidability reduction combines an effective Davis–Putnam–Robinson–Matiyasevich representation with a bounded-order edge–triangle construction adapted from Hatami and Norine.

Both results follow from an effective finite compiler theorem that converts bounded local optimization with a fixed sampling order over finite weighted binary matrices into exact Turán densities of finite forbidden families. For each fixed sampling scheme, target uniformity, and common denominator, the resulting density is $A+Bt$, where t is the optimization value and the constants $A \in \mathbb{Q}$ and $B \in \mathbb{Q}_{>0}$ are effectively computable and independent of the coefficient vector. Via finite sampling, the theorem also covers bounded local functionals on ordinary graphons. As an application, for every finite bipartite graph H , the Sidorenko property of H is equivalent to equality between the Turán densities of two effectively constructed finite forbidden families.

1 Introduction

For an integer $r \geq 2$, an r -graph is a family of r -subsets of a finite vertex set. Given a finite family $\mathcal{F}$ of r -graphs, an r -graph is $\mathcal{F}$ -free if it contains no member of $\mathcal{F}$ as a subgraph; copies need not be induced. Let $\text{ex}(n, \mathcal{F})$ be the maximum number of edges in an n -vertex $\mathcal{F}$ -free r -graph. We use the conventions $\text{ex}(n, \emptyset) := \binom{n}{r}$ and $\text{ex}(n, \mathcal{F}) := 0$ if no n -vertex $\mathcal{F}$ -free r -graph exists. The Turán density of $\mathcal{F}$ is $\pi(\mathcal{F}) := \lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F}) / \binom{n}{r}$. This limit exists by the averaging argument of Katona, Nemetz, and Simonovits [23]. For a single r -graph H , write $\pi(H) := \pi(\{H\})$.

At the level of densities, the graph case is completely understood. Turán’s theorem [44] determines $\text{ex}(n, K_t)$, while the Erdős–Stone–Simonovits theorem [11, 10] determines $\pi(\mathcal{F})$ from $\min_{F \in \mathcal{F}} \chi(F)$ whenever $\mathcal{F}$ is nonempty and each member has an edge. Together with the preceding conventions, this makes $\pi(\mathcal{F})$ rational and computable when $r = 2$.

The situation changes sharply for $r \geq 3$. Even the density of the tetrahedron $K_4^{(3)}$ is unknown: Turán [44] conjectured in 1941 that $\pi(K_4^{(3)}) = 5/9$. This is the classical tetrahedron problem, or Turán’s $(3, 4)$ -problem. Flag-algebra and semidefinite methods for this and related small

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3-graph problems were developed by Razborov [40], Baber and Talbot [2], and Falgas-Ravry and Vaughan [12]; see also the surveys of Füredi [17], Sidorenko [42], and Keevash [24]. Thus even very small forbidden hypergraphs may have densities that are difficult to determine.

The hypergraph Turán problem has both a local and a global aspect. For a specific forbidden family, one seeks the extremal constructions, their stability, and the value of the limiting density. From a broader viewpoint, one asks which numbers can arise as Turán densities, what structure the set of all such densities has, and how much information about $\pi(\mathcal{F})$ can be recovered from the finite description of $\mathcal{F}$. These questions measure the extent to which a finite local prohibition controls a global asymptotic optimum. For graphs, chromatic number provides a complete organizing principle at the level of densities; for hypergraphs, no comparable general invariant is known.

The main contribution of this paper is to broaden this general picture in two complementary directions. On the algorithmic side, we show that, for every sufficiently large fixed uniformity, even comparison with a rational threshold is undecidable. On the arithmetic side, we construct finite forbidden families with transcendental Turán density. A common compiler theorem underlies both results: it embeds bounded local optimization over finite weighted binary structures into Turán densities while preserving the optimization value up to a strictly increasing rational affine map. Thus a finite local prohibition can encode both an undecidable decision problem and an asymptotic optimum outside the algebraic numbers.

We begin with the algorithmic direction. Even when exact evaluation is out of reach, one may ask whether the density can at least be compared with a prescribed rational threshold. This comparison is inherently infinitary: although $\mathcal{F}$ is finite, $\pi(\mathcal{F})$ is defined through $\mathcal{F}$ -free hypergraphs of unbounded order. No finite decision criterion analogous to the graph case is known. In a related setting, Hatami and Norine [21] proved that it is undecidable whether a given linear inequality between graph homomorphism densities holds for every graph. Their result does not concern Turán densities of finite forbidden families, but the graphon, homomorphism-density, and quantum-graph machinery behind it [29, 16, 28] provides the local-density input for our construction; Razborov’s flag algebras [39] provide closely related finite-sampling tools. Pikhurko [37, Question 29] asked whether, for a fixed $r \geq 3$, one can decide, given $\mathcal{F}$ and a rational number α , whether $\pi(\mathcal{F}) \leq \alpha$. Our first theorem answers this question negatively for every sufficiently large uniformity.

We fix standard encodings of all finite objects. A map or construction is *computable*, *effective*, or *effectively computable* if an algorithm takes the encodings of its stated inputs and halts with the asserted finite output. A fixed constant is effectively computable if an algorithm halts with its value. In particular, all uniformity thresholds and forbidden families below are produced algorithmically.

Theorem 1.1. *There exists an integer $r_0^{\text{rat}} \geq 3$ such that, for every fixed $r \geq r_0^{\text{rat}}$, each of the following decision problems is undecidable:*

- (i) *given a finite family $\mathcal{F}$ of r -graphs and a rational number α , determine whether $\pi(\mathcal{F}) \leq \alpha$;*
- (ii) *given two finite families $\mathcal{F}$ and $\mathcal{G}$ of r -graphs, determine whether $\pi(\mathcal{F}) \leq \pi(\mathcal{G})$.*

The reductions prove a little more. The two problems in Theorem 1.1 remain undecidable under the respective promises $\pi(\mathcal{F}) \geq \alpha$ and $\pi(\mathcal{F}) \geq \pi(\mathcal{G})$. Under these promises, $\leq$ is equivalent to equality and $>$ is its complement, so the corresponding equality and strict-inequality problems are also undecidable.

A second basic question concerns the range of possible Turán densities. For $r \geq 3$, let $\Pi_{\text{fin}}^{(r)}$ be the set of densities of finite forbidden families and let $\Pi_{\infty}^{(r)}$ be the corresponding set when the forbidden family may be arbitrary. Erdős [9] initiated the study of gaps in these sets. He proved that $\Pi_{\infty}^{(r)} \cap (0, r!/r^r) = \emptyset$ and asked whether every $\alpha < 1$ is a *jump*, that is, whether

$\Pi_\infty^{(r)} \cap (\alpha, \alpha + c) = \emptyset$ for some $c > 0$. Frankl and Rödl [15] disproved this by constructing non-jumps in every uniformity; see also Baber and Talbot [1] and Pikhurko [38] for explicit jumps and quantitative results on gaps.

Brown and Simonovits [5] proved that the density of an arbitrary forbidden family is the limit of the densities of its finite subfamilies. Pikhurko [37, Corollary 4 and Theorem 24] then identified $\Pi_\infty^{(r)}$ with $\overline{\Pi_{\text{fin}}^{(r)}}$ and showed that this closed set has cardinality continuum. Frankl, Peng, Rödl, and Talbot [14] asked whether $\Pi_\infty^{(r)}$ contains a terminal interval. Liu and Pikhurko [27] recently answered this affirmatively: for every $r \geq 3$, some $\delta_r > 0$ satisfies $[1 - \delta_r, 1] \subseteq \Pi_\infty^{(r)}$. Consequently, $\Pi_\infty^{(r)}$ has Hausdorff dimension 1, and $\Pi_{\text{fin}}^{(r)}$ is dense in a terminal interval, although this countable set cannot itself contain a nondegenerate interval.

For finite forbidden families, Baber and Talbot [2] and Pikhurko [37, Theorem 5] independently found irrational algebraic Turán densities; Pikhurko’s construction works in every uniformity $r \geq 3$. Grosu [19] initiated a systematic study of the algebraic and topological structure of Turán densities. Yan and Peng [46] and Wu [45] later constructed single hypergraphs with irrational density, and Liu and Pikhurko [26] proved that the algebraic degrees of finite-family Turán densities are unbounded in every uniformity $r \geq 3$. These results naturally lead to the next question: can the density of a finite forbidden family be transcendental?

Transcendental constants were previously known in related extremal models. Mubayi and Terry [36] proved that a locally constrained multigraph product maximum is $2^{\gamma n^2 + O(n)}$ for an explicit transcendental constant γ ; transcendence of the associated enumeration base is conditional on Schanuel’s conjecture. Yuster [47] proved unconditionally that the inducibility of P_4 in ordered monotone balanced bipartite graphs is the transcendental number $2/e^2$. Neither result is a Turán density of a finite forbidden hypergraph family. At the 2011 AIM workshop, Fox asked whether such a density can be transcendental; see Mubayi, Pikhurko, and Sudakov [35, Section 3, Question 2] and Pikhurko [37, Question 27]. We answer this question affirmatively for every sufficiently large uniformity.

Theorem 1.2. *For every sufficiently large integer r , there is a finite family of r -graphs with transcendental Turán density.*

Both theorems will follow from the compiler theorem stated below. At the level of a single finite r -graph G , Pikhurko [37, Corollary 4 and Theorem 24] proved that $r!\lambda(G)$ is the Turán density of a finite forbidden family. Local sampling problems require more: their optima may be approached only by weighted structures of increasing order and need not be represented by the Lagrangian of any one finite hypergraph. Our compiler encodes the entire sampling problem in a finitely based monotone class, proves that its class Lagrangian is a strictly increasing rational affine function of the sampling value, and then realizes this class Lagrangian as a Turán density.

Although the language resembles classical matrix-pattern avoidance [18, 30], the optimization problem is different. Rather than maximizing the number of entries subject to avoiding a deterministic submatrix, we optimize the expected coefficient of a bounded induced pattern sampled from a probability-weighted binary relation. We now introduce the finite sampling language needed to state the compiler theorem.

A *finite binary matrix sampling scheme* is a triple $\mathfrak{M} = (I, \mathbf{d}, \mathcal{R})$, where I is a nonempty finite ordered set of sorts, $\mathbf{d} = (d_i)_{i \in I}$ gives a positive sample multiplicity for each sort, and $\mathcal{R}$ specifies the visible diagonal and off-diagonal relation blocks. We require $d_i \geq 2$ whenever $\{i, i\} \in \mathcal{R}$, and call $d(\mathfrak{M}) := \sum_{i \in I} d_i$ the *total sampled order*.

Intuitively, a scheme describes a bounded local binary experiment. Each sort is a vertex population, d_i records how many positions are sampled from it, and $\mathcal{R}$ specifies which within-population and cross-population entries are observed. One diagonal sort gives ordinary graph sampling, while two sorts with only their cross-block visible give rectangular matrix sampling.

Allowing several sorts makes it possible to vary several local structures independently and to multiply their statistics. This is essential for the undecidability reduction, where different sorts encode the finitely many variables in a fixed Diophantine representation and an auxiliary sort supplies the required penalty factor. Because that representation has a fixed number of variables and bounded degree, the total sampled order remains fixed.

A *finite weighted $\mathfrak{M}$ -graph* consists of a finite weighted vertex set in each sort and a 0–1 relation on every visible block, with the weights in each sort summing to one. Sampling d_i vertices independently with replacement from every sort i produces an induced pattern. Let $\mathcal{U}_{\mathfrak{M}}$ be the finite set of isomorphism classes of these patterns, and let $\mathbf{I}_{\mathfrak{M}}(\beta)$ denote the induced-density basis element associated with $\beta \in \mathcal{U}_{\mathfrak{M}}$.

An *input* is a coefficient vector $(a_{\beta})_{\beta \in \mathcal{U}_{\mathfrak{M}}}$, represented by $\mathbf{f} := \sum_{\beta \in \mathcal{U}_{\mathfrak{M}}} a_{\beta} \mathbf{I}_{\mathfrak{M}}(\beta)$. We use rational coefficients in $[0, 1]$. The *matrix value* $\text{Val}_{\mathfrak{M}}(\mathbf{f})$ is the supremum, over all finite weighted $\mathfrak{M}$ -graphs, of the expected coefficient of the sampled pattern; the formal definitions are given in [Section 2.1](#). A positive integer N is a *common denominator* if $a_{\beta} \in N^{-1}\mathbb{Z}$ for every β .

The compiler theorem turns this finite sampling problem into a Turán-density problem while preserving its value up to a strictly increasing rational affine map.

Theorem 1.3. *For every integer $d \geq 2$, one can effectively compute an integer $r_{\text{aff}}(d)$ such that the following holds. Let $\mathfrak{M}$ be a finite binary matrix sampling scheme of total sampled order d , let $r \geq r_{\text{aff}}(d)$, and let N be a positive integer. There are effectively computable numbers $A \in \mathbb{Q}$ and $B \in \mathbb{Q}_{>0}$, depending only on $\mathfrak{M}$, r , and N , such that, for every input $\mathbf{f}$ whose coefficients lie in $N^{-1}\mathbb{Z} \cap [0, 1]$, one can effectively construct a finite family $\mathcal{F}$ of r -graphs satisfying $\pi(\mathcal{F}) = A + B \text{Val}_{\mathfrak{M}}(\mathbf{f})$.*

The compiler uses only the finite input data—the scheme, target uniformity, common denominator, and coefficient vector—and never the generally unknown value $\text{Val}_{\mathfrak{M}}(\mathbf{f})$. Two uniformity features are crucial. First, $r_{\text{aff}}(d)$ depends only on the total sampled order, so the target uniformity can be fixed before the common denominator is known. Second, once $\mathfrak{M}$, r , and N are fixed, the same strictly increasing rational affine map $t \mapsto A + Bt$ applies to every coefficient vector.

Ordinary graphon optimization is the one-sort diagonal specialization of [Theorem 1.3](#). Let $\mathcal{U}_d$ be the set of isomorphism classes of graphs on d vertices, and let $W : [0, 1]^2 \rightarrow [0, 1]$ be a graphon. Sample independent uniform points $X_1, \dots, X_d$ and, conditional on them, include each pair $uv \in \binom{[d]}{2}$ independently with probability $W(X_u, X_v)$. Denote the resulting random graph on $[d]$ by $\mathbb{G}(d, W)$. For $a = (a_{\beta})_{\beta \in \mathcal{U}_d}$, define

$$\text{Val}_d^{\text{gr}}(a) := \sup_W \mathbb{E} a_{[\mathbb{G}(d, W)]},$$

where the supremum ranges over all graphons.

Applying the compiler theorem to this one-sort scheme gives its ordinary graphon specialization.

Corollary 1.4. *Let $d \geq 2$, let $r \geq r_{\text{aff}}(d)$, and let N be a positive integer. There are effectively computable numbers $A \in \mathbb{Q}$ and $B \in \mathbb{Q}_{>0}$, depending only on d , r , and N , such that every coefficient vector $a = (a_{\beta})_{\beta \in \mathcal{U}_d}$ with $a_{\beta} \in N^{-1}\mathbb{Z} \cap [0, 1]$ yields an effectively constructible finite family $\mathcal{F}$ of r -graphs satisfying $\pi(\mathcal{F}) = A + B \text{Val}_d^{\text{gr}}(a)$.*

This is an exact statement for rational functionals determined by one fixed induced graph sample. Its proof is given in [Section 2.2](#); more general bounded local graphon functionals enter through standard finite-sampling approximation.

The graphon specialization also gives a direct equality reduction for Sidorenko’s conjecture [\[41\]](#); see also [\[20, 6, 7, 25\]](#). Let H be a finite bipartite graph. For a graphon W , put

$$p(W) := t(K_2, W) = \int_{[0, 1]^2} W(x, y) dx dy$$

and

$$t(H, W) := \int_{[0,1]^{V(H)}} \prod_{uv \in E(H)} W(x_u, x_v) \prod_{u \in V(H)} dx_u.$$

We use the convention $p(W)^0 := 1$. We say that H has the *Sidorenko property* if, for every graphon W ,

$$t(H, W) \geq p(W)^{e(H)}. \quad (1)$$

Equivalently, the gap

$$\Delta_H^{\text{gr}} := \sup_W (p(W)^{e(H)} - t(H, W)),$$

where the supremum ranges over all graphons, is zero. The normalized functional

$$W \mapsto \frac{1}{2} + \frac{1}{2} (p(W)^{e(H)} - t(H, W))$$

depends only on a finite graph sample: use the vertices of H together with a vertex-disjoint matching of size $e(H)$ at one common sampling order. Its supremum is $1/2 + \Delta_H^{\text{gr}}/2$, so [Corollary 1.4](#) yields the following.

Corollary 1.5. *There is an effectively computable function $r_{\text{Sid}}: \mathbb{N} \rightarrow \mathbb{N}$ with the following property. Given a finite bipartite graph H on $n \geq 1$ vertices and an integer $r \geq r_{\text{Sid}}(n)$, one can effectively construct two finite families $\mathcal{F}_{H,r}$ and $\mathcal{F}_{H,r}^0$ of r -graphs such that*

$$\pi(\mathcal{F}_{H,r}) = \pi(\mathcal{F}_{H,r}^0) \iff H \text{ has the Sidorenko property.}$$

Thus the compiler expresses Sidorenko's conjecture as equality between the Turán densities of two effectively constructed finite families.

Proof sketch of Theorem 1.3. Fix $\mathfrak{M}$, r , N , and an input $\mathfrak{f}$, and put $t := \text{Val}_{\mathfrak{M}}(\mathfrak{f})$. The proof follows the scheme

$$\mathfrak{f} \xrightarrow{\text{finite encoding}} \mathcal{C}_{\mathfrak{f}} \xrightarrow{\text{affine evaluation}} \Lambda(\mathcal{C}_{\mathfrak{f}}) \xrightarrow{\text{pair-covering transfer}} \pi(\mathcal{F}_{\mathfrak{f}}).$$

For the first step, a finite collection of rooted templates records the sampled binary patterns and their coefficients. The r -graphs admitting a consistent partial record form a monotone class $\mathcal{C}_{\mathfrak{f}}$. Every failure of consistency has a bounded witness, so the roots can be eliminated effectively and $\mathcal{C}_{\mathfrak{f}}$ can be described by a finite obstruction family.

The main step is to evaluate the class Lagrangian. The templates are weighted so that an optimizing configuration can be balanced and symmetrized without decreasing its weighted edge polynomial. Its remaining local data then form a finite weighted $\mathfrak{M}$ -graph, whose contribution is bounded by t ; uniformly weighted finite $\mathfrak{M}$ -graphs give the matching lower bound. The parameters are chosen so that the same balanced point is optimal for every $t \in [0, 1]$. Consequently,

$$\Lambda(\mathcal{C}_{\mathfrak{f}}) = h_{\mathfrak{M},r,N} + b_{\mathfrak{M},r,N}t,$$

where $h_{\mathfrak{M},r,N} \in \mathbb{Q}$ and $b_{\mathfrak{M},r,N} \in \mathbb{Q}_{>0}$ depend only on $\mathfrak{M}$, r , and N .

Finally, bounded pair-covering extensions of the obstructions define a finite family $\mathcal{F}_{\mathfrak{f}}$. Blowups of members of $\mathcal{C}_{\mathfrak{f}}$ give one inequality in

$$\pi(\mathcal{F}_{\mathfrak{f}}) = r! \Lambda(\mathcal{C}_{\mathfrak{f}}),$$

while a minimum-support Lagrangian-maximizing weighting gives the reverse inequality. Combining the last two displays yields the required rational affine formula.

The rest of the paper is organized as follows. [Section 2](#) develops the basic definitions and finite-sampling tools. [Section 3](#) reduces the compiler theorem to finite encoding, affine Lagrangian evaluation, and pair-covering transfer. The first two ingredients are proved in [Sections 4](#) and [5](#). Finally, [Section 6](#) proves the Sidorenko consequence and the two main theorems.

2 Preliminaries

This section fixes the notation and finite-sampling framework used throughout the proof. We first recall hypergraph Lagrangians, blowups, and monotone classes, and then formalize finite binary matrix sampling schemes. The main result of the section identifies sampling with replacement with the supremum of a squarefree sampling polynomial.

For a nonnegative integer m , put $[m] := \{1, \dots, m\}$, with $[0] = \emptyset$, and set $\mathbb{N} := \{1, 2, \dots\}$. For a finite set X , write $\binom{X}{j}$ for its family of j -element subsets. We use $(z)_j := z(z-1) \cdots (z-j+1)$ for the falling factorial, with $(z)_0 := 1$, and $e_j(z_1, \dots, z_m) := \sum_{I \in \binom{[m]}{j}} \prod_{i \in I} z_i$ for the j th elementary symmetric polynomial. Throughout, $\log$ denotes the natural logarithm.

For an r -graph G , let $V(G)$, $E(G)$, $v(G)$, and $e(G)$ denote its vertex set, edge set, order, and size. Unless stated otherwise, subgraphs need not be induced; $G[X]$ is the subgraph induced by $X \subseteq V(G)$, and $[G]$ is the isomorphism class of G .

All algorithmic statements use the finite encodings fixed in the Introduction. A set is *computably enumerable* if an algorithm lists all its elements; equivalently, an algorithm halts and accepts precisely on the members of the set, with no requirement that it halt on nonmembers. For a *promise problem*, an algorithm must halt and answer correctly on every input satisfying the promise, with no requirement on other inputs.

For a family $\mathcal{B}$ of r -graphs, let $\text{Forb}(\mathcal{B})$ be the class of finite r -graphs containing no copy of a member of $\mathcal{B}$. If $\mathcal{C} = \text{Forb}(\mathcal{B})$ and $\mathcal{B}$ is finite, then $\mathcal{B}$ is a finite *forbidden family*, or *obstruction family*, for $\mathcal{C}$. A class is *monotone* if it is closed under taking subgraphs.

We use the following normalization of hypergraph Lagrangians. For an r -graph G with $V(G) \neq \emptyset$, a *weighting* is a vector $x = (x_v)_{v \in V(G)}$ with $x_v \geq 0$ for every v and $\sum_v x_v = 1$. Its *support* is $\text{supp}(x) := \{v : x_v > 0\}$. The *Lagrangian polynomial* is

$$p_G(x) := \sum_{e \in E(G)} \prod_{v \in e} x_v.$$

The *Lagrangian* of G is

$$\lambda(G) := \max\{p_G(x) : x \text{ is a weighting of } G\}.$$

A *minimum-support maximizing weighting* is a maximizer whose support has minimum cardinality. For any class $\mathcal{C}$ containing an r -graph with at least one vertex, define its *class Lagrangian* by $\Lambda(\mathcal{C}) := \sup\{\lambda(G) : G \in \mathcal{C}, v(G) > 0\}$.

A *blowup* of G replaces every vertex by an independent cluster and every edge by the complete r -partite r -graph across the corresponding clusters. If the cluster proportions converge to a weighting x while the orders tend to infinity, then the edge densities converge to $r!p_G(x)$. Thus the supremum of the limiting densities of blowups of G is $r!\lambda(G)$. This normalization accounts for the factor $r!$ in the proof of [Theorem 1.3](#). The weighting method originates in the graph theorem of Motzkin and Straus [\[34\]](#) and was developed for hypergraphs in [\[15, 13, 43\]](#); for the role of blowups in extremal densities, see Brown and Simonovits [\[5\]](#). Pikhurko [\[37, Corollary 4 and Theorem 24\]](#) proved that $r!\lambda(G)$ is the Turán density of a finite family for every finite r -graph G , and that these values are dense in $\Pi_\infty^{(r)}$ when $r \geq 3$.

2.1 Finite matrix sampling schemes

We now formalize the finite binary matrix sampling schemes introduced in the Introduction. After defining their induced-density basis and sampling evaluation, [Lemma 2.1](#) identifies the matrix value of a nonnegative input with the supremum of a squarefree polynomial. It also

supplies uniformly weighted finite objects approaching that supremum, which are the objects represented by the later hypergraph construction.

A *finite binary matrix sampling scheme* consists of a nonempty finite ordered set I of sorts, a tuple $\mathbf{d} = (d_i)_{i \in I}$ of positive integers, and a set $\mathcal{R}$ of unordered pairs $\{i, j\}$ of sorts, where $i = j$ is allowed. Write $\mathfrak{M} := (I, \mathbf{d}, \mathcal{R})$. The pairs in $\mathcal{R}$ are the *visible blocks*; we require $d_i \geq 2$ whenever $\{i, i\} \in \mathcal{R}$. Set $d(\mathfrak{M}) := \sum_{i \in I} d_i$ and $\nu_{\mathfrak{M}} := \prod_{i \in I} d_i!$.

An $\mathfrak{M}$ -graph A has a vertex partition $V(A) = \bigsqcup_{i \in I} V_i(A)$, a within-sort relation $E_{ii}(A) \subseteq \binom{V_i(A)}{2}$ for every visible diagonal block, and a cross relation $E_{ij}(A) \subseteq V_i(A) \times V_j(A)$ for every visible block $\{i, j\}$ with $i < j$. Isomorphisms preserve all sorts and relations; write $[A]_{\mathfrak{M}}$ for the isomorphism class and $\text{Aut}_{\mathfrak{M}}(A)$ for the automorphism group.

For two $\mathfrak{M}$ -graphs on the same sorted vertex sets, write $A \subseteq A'$ if every relation of A is contained in the corresponding relation of A' . Let $|E(A)|$ be the total number of entries in all relations, and for $X_i \subseteq V_i(A)$ write $A[\bigcup_i X_i]$ for the induced $\mathfrak{M}$ -subgraph on these sorted vertex sets. Let $\mathcal{U}_{\mathfrak{M}}$ be the set of isomorphism classes of $\mathfrak{M}$ -graphs with exactly d_i vertices of sort i for every i . An $\mathfrak{M}$ -quantum graph is a finite real linear combination of isomorphism classes of $\mathfrak{M}$ -graphs.

For $\beta = [A]_{\mathfrak{M}} \in \mathcal{U}_{\mathfrak{M}}$, the *induced expansion* is defined by

$$\text{ind}_{\mathfrak{M}}(A) := \sum_{A' \supseteq A} (-1)^{|E(A')| - |E(A)|} A',$$

where A' ranges over all supergraphs on the same sorted vertex sets using only blocks in $\mathcal{R}$, after which vertex names are forgotten. Define $\mathbf{I}_{\mathfrak{M}}(\beta) := (\nu_{\mathfrak{M}} / |\text{Aut}_{\mathfrak{M}}(A)|) \text{ind}_{\mathfrak{M}}(A)$. This does not depend on the representative A . Ordering the finite patterns by their numbers of relation entries, Boolean Möbius inversion gives a triangular change-of-basis matrix with nonzero diagonal. Hence the elements $\mathbf{I}_{\mathfrak{M}}(\beta)$ form a basis of the subspace spanned by the patterns in $\mathcal{U}_{\mathfrak{M}}$, called the *$\mathfrak{M}$ -induced-density basis*.

The factor $\nu_{\mathfrak{M}} / |\text{Aut}_{\mathfrak{M}}(A)|$ gives the probabilistic normalization: it is the number of distinct sorted labelings of A on the prescribed sampled positions. More generally, for $\mathfrak{M}$ -graphs F and H on the sampled sort sizes, let $\text{inj}_{\mathfrak{M}}(F, H)$ be the number of sort-preserving, edge-preserving bijections from F to a representative of H . Boolean Möbius inversion, grouped by sorted isomorphism class, gives

$$F = \sum_{H \in \mathcal{U}_{\mathfrak{M}}} \frac{\text{inj}_{\mathfrak{M}}(F, H)}{\nu_{\mathfrak{M}}} \mathbf{I}_{\mathfrak{M}}(H). \quad (2)$$

Indeed, the labeled identity $F = \sum_{A' \supseteq F} \text{ind}_{\mathfrak{M}}(A')$ follows from Boolean inversion, and grouping its terms gives the displayed coefficient. Consequently, evaluating $\mathbf{I}_{\mathfrak{M}}(\beta)$ records the probability that the sampled pattern has isomorphism class β ; the basis elements partition the sampling outcomes.

An *input* is an $\mathfrak{M}$ -quantum graph expressed in this basis. We write $\mathbf{f} := \sum_{\beta \in \mathcal{U}_{\mathfrak{M}}} a_{\beta} \mathbf{I}_{\mathfrak{M}}(\beta)$ and call the numbers a_{β} its *induced-density coefficients*.

A *finite weighted $\mathfrak{M}$ -graph* is a finite $\mathfrak{M}$ -graph G with every sort nonempty and with a probability weighting $x^{(i)}$ on each $V_i(G)$. Write $\mathbf{x} := (x^{(i)})_{i \in I}$. Independently for each sort i , sample a map $\phi_i: [d_i] \rightarrow V_i(G)$ with replacement according to $x^{(i)}$. Pulling back the relations of G gives a random pattern $\mathbb{M}_{\mathfrak{M}}(G, \phi)$ on the fixed sampled positions, where $\phi := (\phi_i)_{i \in I}$; a repeated image in a diagonal block is treated as a nonedge. Define the *sampling evaluation* and *matrix value* by

$$\hat{t}_{\mathfrak{M}}(\mathbf{f}; G, \mathbf{x}) := \mathbb{E} a_{[\mathbb{M}_{\mathfrak{M}}(G, \phi)]_{\mathfrak{M}}} \quad \text{and} \quad \text{Val}_{\mathfrak{M}}(\mathbf{f}) := \sup_{G, \mathbf{x}} \hat{t}_{\mathfrak{M}}(\mathbf{f}; G, \mathbf{x}).$$

The corresponding squarefree sampling polynomial is

$$T_{\mathbf{f}}^{\mathfrak{M}}(G, \mathbf{x}) := \sum_{(X_i)_{i \in I}} a_{[G[\bigcup_i X_i]]_{\mathfrak{M}}} \prod_{i \in I} \prod_{v \in X_i} x_v^{(i)},$$

where the sum ranges over all tuples $(X_i)_{i \in I}$ with $X_i \in \binom{V_i(G)}{d_i}$ for every i . We write unif for the tuple of uniform sort weights.

For an injective sample, specifying the maps ϕ_i is equivalent to choosing the image sets X_i and then ordering each X_i in $d_i!$ ways. Thus each tuple $(X_i)_{i \in I}$ occurs exactly $\nu_{\mathfrak{M}}$ times. Repeated samples are the only difference between the sampling evaluation and its squarefree part. Nonnegative coefficients allow them to be discarded, and large uniform blowups make their contribution vanish.

Lemma 2.1. *If $\mathfrak{M}$ is a finite binary matrix sampling scheme and the input $\mathfrak{f} = \sum_{\beta \in \mathcal{U}_{\mathfrak{M}}} a_{\beta} \mathbf{I}_{\mathfrak{M}}(\beta)$ has nonnegative induced-density coefficients, then*

$$\text{Val}_{\mathfrak{M}}(\mathfrak{f}) = \nu_{\mathfrak{M}} \sup_{G, \mathbf{x}} T_{\mathfrak{f}}^{\mathfrak{M}}(G, \mathbf{x}).$$

Furthermore, there is a sequence (G_n) of finite $\mathfrak{M}$ -graphs with uniform weights on every sort and $|V_i(G_n)| \rightarrow \infty$ for every $i \in I$, such that $\nu_{\mathfrak{M}} T_{\mathfrak{f}}^{\mathfrak{M}}(G_n, \text{unif}) \rightarrow \text{Val}_{\mathfrak{M}}(\mathfrak{f})$.

Proof. For a fixed weighted $\mathfrak{M}$ -graph, the injective samples contribute exactly $\nu_{\mathfrak{M}} T_{\mathfrak{f}}^{\mathfrak{M}}(G, \mathbf{x})$. All other contributions are nonnegative, so $\text{Val}_{\mathfrak{M}}(\mathfrak{f}) \geq \nu_{\mathfrak{M}} \sup_{G, \mathbf{x}} T_{\mathfrak{f}}^{\mathfrak{M}}(G, \mathbf{x})$.

For the reverse inequality, choose $(G_m, \mathbf{x}_m)$ with $\widehat{t}_{\mathfrak{M}}(\mathfrak{f}; G_m, \mathbf{x}_m) > \text{Val}_{\mathfrak{M}}(\mathfrak{f}) - 1/m$. Since the evaluation is polynomial in the sortwise weights, we may make all weights rational at a cost of less than $1/m$ in value. Delete the zero-weight vertices and replace each remaining vertex by a positive number of twins proportional to its weight, pulling back every relation. Distinct twins of the same vertex are nonadjacent in a diagonal block, so sampling them has the same outcome as repeating the original vertex. Thus uniform weighting on the blowup preserves the sampling evaluation.

Multiplying all twin multiplicities by a common factor still preserves the evaluation and makes every sort arbitrarily large. Choose the factor so that every sort has at least m vertices and the following union bound is less than $1/m$. If $\tilde{n}_i$ is the resulting size of sort i , then

$$\Pr(\text{some sampled vertex is repeated}) \leq \sum_{i \in I} \binom{d_i}{2} \frac{1}{\tilde{n}_i} < \frac{1}{m}.$$

Let $\tilde{G}_m$ be this uniformly weighted blowup and put $a_* := \max_{\beta} a_{\beta}$. Since the noninjective samples contribute at most a_*/m , we obtain

$$\text{Val}_{\mathfrak{M}}(\mathfrak{f}) - \frac{2 + a_*}{m} < \nu_{\mathfrak{M}} T_{\mathfrak{f}}^{\mathfrak{M}}(\tilde{G}_m, \text{unif}) \leq \text{Val}_{\mathfrak{M}}(\mathfrak{f}).$$

The right inequality was proved above. Letting $m \rightarrow \infty$ gives the reverse inequality and the required uniformly weighted sequence. $\square$

We finish with the two specializations used in the applications. For $d \geq 2$, let $\mathfrak{G}_d := (\{1\}, (d), \{\{1, 1\}\})$. Its finite objects are symmetric 0–1 matrices with zero diagonal, equivalently finite graphs, and $\mathcal{U}_{\mathfrak{G}_d}$ is the set of d -vertex graph patterns. If G_A is the graph with adjacency matrix A , then $\widehat{t}_{\mathfrak{G}_d}(\mathfrak{f}; G_A, x)$ is the expected coefficient obtained by sampling d vertices independently according to x , with repetitions treated as nonedges.

For positive integers p and q , order two sorts by $L < R$ and set $\mathfrak{B}_{p,q} := (\{L, R\}, (p, q), \{\{L, R\}\})$. Its finite objects are rectangular 0–1 matrices, and its patterns are binary $p \times q$ matrices modulo independent row and column permutations. This specialization is therefore invariant under separate row and column permutations, unlike the order-preserving matrix-containment model discussed in the Introduction.

2.2 The ordinary graphon specialization

We conclude the preliminaries by proving that the one-sort finite model has exactly the same supremum as ordinary graphon sampling.

Proof of Corollary 1.4. Use the one-sort diagonal scheme $\mathfrak{G}_d$, identify $\mathcal{U}_{\mathfrak{G}_d}$ with $\mathcal{U}_d$, and set $\mathbf{f}_a := \sum_{\beta \in \mathcal{U}_d} a_\beta \mathbf{I}_{\mathfrak{G}_d}(\beta)$. Every finite weighted graph gives a step graphon: partition $[0, 1]$ into intervals whose measures are the vertex weights, use the corresponding 0–1 adjacency relation between distinct vertex intervals, and set the kernel equal to zero on the square of each vertex interval. The two sampling evaluations are then equal, so

$$\text{Val}_{\mathfrak{G}_d}(\mathbf{f}_a) \leq \text{Val}_d^{\text{gr}}(a).$$

Conversely, fix a graphon W and an integer $m \geq d$. Independently sample m uniform points and, conditional on those points, form a random graph G_m by drawing every edge independently with the probability prescribed by W . Give G_m the uniform weighting, and let $\mathcal{E}_m$ be the event that the d vertices in its sampling evaluation are distinct. Then

$$\Pr(\mathcal{E}_m) = \frac{\binom{m}{d}}{m^d} \rightarrow 1.$$

Conditional on $\mathcal{E}_m$, averaging also over the construction of G_m gives exactly the distribution of $\mathbb{G}(d, W)$. Since every input coefficient lies in $[0, 1]$,

$$\mathbb{E} \hat{t}_{\mathfrak{G}_d}(\mathbf{f}_a; G_m, \text{unif}) \geq \Pr(\mathcal{E}_m) \mathbb{E} a_{[\mathbb{G}(d, W)]}.$$

Some realization of G_m attains at least the expectation. Letting $m \rightarrow \infty$ and then taking the supremum over W proves the reverse inequality. Hence

$$\text{Val}_{\mathfrak{G}_d}(\mathbf{f}_a) = \text{Val}_d^{\text{gr}}(a).$$

Applying Theorem 1.3 to $\mathfrak{G}_d$ gives the stated formula and effective construction, with constants depending only on d , r , and N . $\square$

3 The compiler reduction

The proof has a construction stage and a transfer stage. Given an input $\mathbf{f}$, the construction produces a monotone class $\mathcal{C}_{\mathbf{f}}$ of r -graphs. The next two lemmas show that this class has an effectively computable finite obstruction family and that its class Lagrangian is a rational affine function of the matrix value with positive slope. The transfer stage then converts this class Lagrangian into the Turán density of a finite family by replacing each obstruction with finitely many pair-covering extensions.

Lemma 3.1. *For every $d \geq 2$, one can effectively compute an integer $r_{\text{aff}}(d)$ with the following property. Let $\mathfrak{M}$ be a finite binary matrix sampling scheme of total sampled order d , let $r \geq r_{\text{aff}}(d)$, and let N be a positive integer. For every input $\mathbf{f} = \sum_{\beta \in \mathcal{U}_{\mathfrak{M}}} a_\beta \mathbf{I}_{\mathfrak{M}}(\beta)$ with $a_\beta \in N^{-1}\mathbb{Z} \cap [0, 1]$ for every $\beta \in \mathcal{U}_{\mathfrak{M}}$, one can effectively construct a monotone class $\mathcal{C}_{\mathbf{f}}$ and a finite family $\mathcal{A}_{\mathbf{f}}$ of r -graphs satisfying*

$$\mathcal{C}_{\mathbf{f}} = \text{Forb}(\mathcal{A}_{\mathbf{f}}).$$

Moreover, the class $\mathcal{C}_{\mathbf{f}}$ contains an r -graph with an edge.

Lemma 3.2. *Let $d \geq 2$, let $\mathfrak{M}$ be a finite binary matrix sampling scheme of total sampled order d , let $r \geq r_{\text{aff}}(d)$, and let N be a positive integer. There are effectively computable*

numbers $h_{\mathfrak{M},r,N} \in \mathbb{Q}$ and $b_{\mathfrak{M},r,N} \in \mathbb{Q}_{>0}$, depending only on $\mathfrak{M}$, r , and N , such that every input $\mathfrak{f} = \sum_{\beta \in \mathcal{U}_{\mathfrak{M}}} a_{\beta} \mathbf{I}_{\mathfrak{M}}(\beta)$ with $a_{\beta} \in N^{-1}\mathbb{Z} \cap [0, 1]$ for every $\beta \in \mathcal{U}_{\mathfrak{M}}$ satisfies

$$\Lambda(\mathcal{C}_{\mathfrak{f}}) = h_{\mathfrak{M},r,N} + b_{\mathfrak{M},r,N} \text{Val}_{\mathfrak{M}}(\mathfrak{f}), \quad (3)$$

where $\mathcal{C}_{\mathfrak{f}}$ is the class given by [Lemma 3.1](#).

The two lemmas are proved in [Sections 4](#) and [5](#), respectively. The first of these sections develops the construction and the required balancing and parameter estimates; the second uses those estimates to evaluate its class Lagrangian. We now show how the lemmas imply [Theorem 1.3](#).

An r -graph *covers pairs* if every pair of its vertices lies in an edge. For an r -graph J , a J -pair-covering extension is an r -graph Q containing a copy of J whose vertex set is pair-covered in Q . More precisely, there is an injection $\iota: V(J) \rightarrow V(Q)$ that maps edges of J to edges of Q and for which every pair in $\iota(V(J))$ lies in an edge of Q .

Such extensions always exist. For every pair of vertices of J not already covered, introduce $r - 2$ new vertices used only for that pair and add the resulting edge. This produces an extension on at most $M(J) := v(J) + (r - 2)\binom{v(J)}{2}$ vertices. Let $\text{Ext}(J)$ contain one representative of each isomorphism class of J -pair-covering extensions on at most $M(J)$ vertices. For every chosen representative Q , fix one witnessing injection $\iota_{J,Q}: V(J) \rightarrow V(Q)$. The injection is used only in the proof below and is forgotten when Q is placed in a forbidden family. The finite family $\text{Ext}(J)$ can be enumerated effectively from J .

The following transfer principle isolates the passage from a finitely based monotone class to a Turán-density problem.

Lemma 3.3 (Pair-covering transfer). *Let $\mathcal{A}$ be a finite family of r -graphs, put $\mathcal{C} := \text{Forb}(\mathcal{A})$, and suppose that $\mathcal{C}$ contains an r -graph with an edge. Define $\mathcal{F} := \bigcup_{J \in \mathcal{A}} \text{Ext}(J)$. Then*

$$\pi(\mathcal{F}) = r! \Lambda(\mathcal{C}). \quad (4)$$

Proof. For the lower bound, fix $G \in \mathcal{C}$ with $v(G) > 0$. Every blowup of G is $\mathcal{F}$ -free. Indeed, suppose that a blowup contained a copy of some $Q \in \text{Ext}(J)$ with $J \in \mathcal{A}$, and transport the fixed image $\iota_{J,Q}(V(J))$ into that copy. Projection onto the blowup clusters is injective on this image: every pair in it lies in an edge, whereas a blowup edge meets each cluster in at most one vertex. The projection would therefore embed J into G , contradicting $G \in \text{Forb}(\mathcal{A})$.

Taking blowups whose cluster proportions approach a maximizing weighting of G gives $\pi(\mathcal{F}) \geq r! \lambda(G)$. Taking the supremum over $G \in \mathcal{C}$ yields

$$\pi(\mathcal{F}) \geq r! \Lambda(\mathcal{C}). \quad (5)$$

For the reverse inequality, let H be an n -vertex $\mathcal{F}$ -free r -graph, where $n \geq r$. We first show that $\lambda(H) \leq \Lambda(\mathcal{C})$. This is immediate if $\lambda(H) = 0$, so assume otherwise. Choose a maximizing weighting x of H with minimum support, and set $D := H[\text{supp}(x)]$. Restricting x to D , or extending a weighting of D by zero, shows that $\lambda(D) = \lambda(H)$.

We claim that D covers pairs. Suppose instead that distinct vertices $u, v \in V(D)$ lie in no common edge of D , and vary x_u, x_v while keeping all other coordinates and the sum $x_u + x_v$ fixed. Any edge of H containing both u and v also contains a vertex outside $\text{supp}(x)$, so its monomial vanishes throughout this line segment. Every remaining nonzero monomial contains at most one of x_u, x_v . Hence p_H is affine on the segment. It attains its maximum at the interior point x , so it must be constant there; both endpoints are then maximizing weightings with smaller support, a contradiction. Thus D covers pairs.

The graph D must belong to $\mathcal{C}$. Indeed, if it contained a copy J' of some $J \in \mathcal{A}$, we could choose, for every pair in $V(J')$, an edge of D containing that pair. The union of J' with these

edges would be a J -pair-covering extension on at most $M(J)$ vertices and hence isomorphic to a member of $\text{Ext}(J)$. This would give a member of $\mathcal{F}$ inside H , a contradiction. Therefore $\lambda(H) = \lambda(D) \leq \Lambda(\mathcal{C})$.

The uniform weighting gives $e(H)n^{-r} \leq \lambda(H)$. Since $n^r/\binom{n}{r} = r! + O_r(n^{-1})$ and $\Lambda(\mathcal{C}) \leq 1/r!$, it follows that

$$\frac{e(H)}{\binom{n}{r}} \leq r!\Lambda(\mathcal{C}) + O_r(n^{-1}).$$

Maximizing over all such H and letting $n \rightarrow \infty$ gives

$$\pi(\mathcal{F}) \leq r!\Lambda(\mathcal{C}). \quad (6)$$

Together with (5), this proves (4). $\square$

Proof of Theorem 1.3. Let $r_{\text{aff}}(d)$ be the threshold from Lemma 3.1, and fix $\mathfrak{M}$, r , and N as in the theorem. Write $h = h_{\mathfrak{M},r,N}$ and $b = b_{\mathfrak{M},r,N}$ for the constants from Lemma 3.2, and set $A_{\mathfrak{M},r,N} := r!h$ and $B_{\mathfrak{M},r,N} := r!b$.

Fix an input $\mathfrak{f}$, and let $\mathcal{C}_{\mathfrak{f}} = \text{Forb}(\mathcal{A}_{\mathfrak{f}})$ be given by Lemma 3.1. Define $\mathcal{F}_{\mathfrak{f}} := \bigcup_{J \in \mathcal{A}_{\mathfrak{f}}} \text{Ext}(J)$. The class $\mathcal{C}_{\mathfrak{f}}$ contains an edge, so Lemma 3.3 and (3) give

$$\pi(\mathcal{F}_{\mathfrak{f}}) = A_{\mathfrak{M},r,N} + B_{\mathfrak{M},r,N} \text{Val}_{\mathfrak{M}}(\mathfrak{f}).$$

The constants belong to $\mathbb{Q}$ and $\mathbb{Q}_{>0}$, respectively, and are independent of the coefficient vector. The finite encoding, affine evaluation, and enumeration of pair-covering extensions are all effective, while the threshold depends only on d . This completes the proof. $\square$

4 Proof of the finite-encoding lemma

We construct the class $\mathcal{C}_{\mathfrak{f}}$ and prove Lemma 3.1. We begin with a finite pair-section palette that turns rational coefficients into densities of root-label cells. These cells are then used to define six families of edge templates. After proving that template representations are unique, we choose the balancing parameters, impose the local consistency conditions, and finally remove the root labels by a bounded-witness argument.

Fix a finite binary matrix sampling scheme $\mathfrak{M} = (I, \mathbf{d}, \mathcal{R})$, and set $d := \sum_{i \in I} d_i \geq 2$ and $\nu_{\mathfrak{M}} := \prod_{i \in I} d_i!$. Throughout this section, $r \geq 4(d+2)$ and N is a positive integer. Fix an input $\mathfrak{f} = \sum_{\beta \in \mathcal{U}_{\mathfrak{M}}} a_{\beta} \mathbf{I}_{\mathfrak{M}}(\beta)$ with $a_{\beta} \in N^{-1}\mathbb{Z} \cap [0, 1]$ for every β , and write $m_{\beta} := Na_{\beta} \in \{0, \dots, N\}$.

All objects below are computable from these data. The uniformity threshold will depend only on d , although the finite label set itself may depend on $\mathfrak{M}$, r , and N .

The order of dependence will be important. The sampled order d alone determines the uniformity threshold $r_{\text{aff}}(d)$. Once $\mathfrak{M}$ and r are fixed, the structural parameters and cell densities are chosen independently of N and of the coefficient vector. The denominator N then enters through the finite label group Ω and its nested parameter sets, while the coefficient vector enters only through the integers $m_{\beta} = Na_{\beta}$. Thus the label set may depend on $\mathfrak{M}$, r , and N , but the target uniformity does not depend on N , and the affine map obtained later is independent of $(a_{\beta})_{\beta}$.

4.1 A finite pair-section palette

By Lemma 2.1, the matrix input enters through a linear combination of squarefree monomials, the same algebraic form as a hypergraph Lagrangian. We encode a rational number m/N as the normalized size of a finite set of root-label tuples.

The next lemma constructs all such sets simultaneously. An index α specifies a later template family, and the parameter m carries the input coefficient. The resulting cell satisfies $|\mathcal{D}_{\alpha,m}|/q^4 =$

$(\theta_\alpha/q^2)(m/N)$. Thus its density is a fixed rational multiple of m/N . Distinct indices occupy distinct cosets, and pair injectivity ensures that, once the index is known, any two coordinates recover the parameters a and ξ . These separation properties distinguish the templates constructed in [Section 4.3](#), while invariance under diagonal translation supplies the symmetry later used to force root-label uniformity.

Lemma 4.1. *Let $\mathbb{C}$ be a finite index set, choose $\theta_\alpha \in \mathbb{Q} \cap [0, 1]$ for every $\alpha \in \mathbb{C}$, and let N and q_0 be positive integers. One can effectively construct a finite product of finite fields $\Omega := \prod_{\gamma \in \mathbb{G}} F_\gamma$, four elements $\lambda_1, \dots, \lambda_4 \in \Omega$, vectors $v_\alpha \in \Omega^4$ for $\alpha \in \mathbb{C}$, and sets $S_{\alpha,m} \subseteq \Omega$ for $\alpha \in \mathbb{C}$ and $m \in \{0, \dots, N\}$, with the following properties. Set $q := |\Omega|$ and define*

$$L := \{(a + \lambda_1 \xi, \dots, a + \lambda_4 \xi) : a \in \Omega \text{ and } \xi \in \Omega\}.$$

The set L is a subgroup of Ω^4 . For $\alpha \in \mathbb{C}$ and $m \in \{0, \dots, N\}$, define

$$\mathcal{D}_{\alpha,m} := v_\alpha + \{(a + \lambda_1 \xi, \dots, a + \lambda_4 \xi) : a \in \Omega \text{ and } \xi \in S_{\alpha,m}\}.$$

- (i) *One has $q \geq q_0$ and $q^2 > |\mathbb{C}|$. The difference $\lambda_i - \lambda_j$ is a unit whenever $i \neq j$. The cosets $v_\alpha + L$ for $\alpha \in \mathbb{C}$ are pairwise distinct.*
- (ii) *For every $\alpha \in \mathbb{C}$, the parameter sets form the nested chain $\emptyset = S_{\alpha,0} \subseteq S_{\alpha,1} \subseteq \dots \subseteq S_{\alpha,N}$, with $|S_{\alpha,m}|/q = \theta_\alpha m/N$ for $0 \leq m \leq N$. Consequently, $|\mathcal{D}_{\alpha,m}|/q^4 = \theta_\alpha m/(q^2 N)$.*
- (iii) *If $\alpha \neq \alpha'$, then $\mathcal{D}_{\alpha,m} \cap \mathcal{D}_{\alpha',m'} = \emptyset$ for all $m, m' \in \{0, \dots, N\}$. Each $\mathcal{D}_{\alpha,m}$ has pair injectivity: its projection onto every pair of coordinates is injective. Every element of $\bigcup_{\alpha \in \mathbb{C}, 0 \leq m \leq N} \mathcal{D}_{\alpha,m}$ determines a unique index α . Once α is known, every pair of coordinates uniquely determines a and ξ in the representation of that element.*
- (iv) *Every $\mathcal{D}_{\alpha,m}$ is invariant under simultaneous diagonal translation $(x_1, x_2, x_3, x_4) \mapsto (x_1 + t, x_2 + t, x_3 + t, x_4 + t)$ for $t \in \Omega$. In each coordinate, the induced action of $(\Omega, +)$ on Ω is regular and therefore transitive.*

Proof. The three factors of Ω will serve separate purposes: one clears the denominators of the fixed multipliers θ_α , one clears the input denominator N , and one makes the product sufficiently large. Express each θ_α in lowest terms and let Q be the least common multiple of their denominators, taking $Q := 1$ if $\mathbb{C} = \emptyset$.

We use the following effective construction. Given a positive integer D , for each prime power $p^e \parallel D$ choose $f(p, e) \geq e$ with $p^{f(p, e)} \geq 4$, and take the product of the fields $\mathbb{F}_{p^{f(p, e)}}$; when $D = 1$, use $\mathbb{F}_5$. The resulting product has order divisible by D , and every factor has at least four elements. The fields and their arithmetic can be constructed effectively, for example by enumerating monic irreducible polynomials.

Apply this construction to Q and N , obtaining products Ω_{int} and Ω_{cof} with $Q \mid |\Omega_{\text{int}}|$ and $N \mid |\Omega_{\text{cof}}|$. Let Ω_{aux} be a product of sufficiently many copies of $\mathbb{F}_5$, and set $\Omega := \Omega_{\text{int}} \times \Omega_{\text{cof}} \times \Omega_{\text{aux}}$. Enlarging the auxiliary factor makes $q = |\Omega|$ at least q_0 and ensures $q^2 > |\mathbb{C}|$, without affecting either divisibility condition.

Choose effective orderings of Ω_{int} and Ω_{cof} . For every α , let $A_\alpha \subseteq \Omega_{\text{int}}$ be the initial segment of size $\theta_\alpha |\Omega_{\text{int}}|$. For $0 \leq m \leq N$, let $B_m \subseteq \Omega_{\text{cof}}$ be the initial segment of size $(m/N) |\Omega_{\text{cof}}|$. Then $B_0 \subseteq B_1 \subseteq \dots \subseteq B_N = \Omega_{\text{cof}}$. Set $S_{\alpha,m} := A_\alpha \times B_m \times \Omega_{\text{aux}}$. The first two factors encode θ_α and m/N separately, and $|S_{\alpha,m}|/|\Omega| = (|A_\alpha|/|\Omega_{\text{int}}|)(|B_m|/|\Omega_{\text{cof}}|) = \theta_\alpha m/N$. This proves the nesting and the parameter-set size assertion in part (ii).

Choose four distinct elements in each field factor and form the corresponding componentwise tuples $\lambda_1, \dots, \lambda_4 \in \Omega$. For $i \neq j$, the difference $\lambda_i - \lambda_j$ is nonzero in every factor and hence is a unit in the direct product.

Consider the additive homomorphism $\varphi: \Omega^2 \rightarrow \Omega^4$ given by $\varphi(a, \xi) = (a + \lambda_1 \xi, \dots, a + \lambda_4 \xi)$. Any two coordinates $i \neq j$ recover $\xi = (\lambda_i - \lambda_j)^{-1}(x_i - x_j)$ and then $a = x_i - \lambda_i \xi$. Hence φ

is injective with image L . Therefore $|L| = q^2$ and Ω^4/L has q^2 cosets. Since $q^2 > |\mathbf{C}|$, distinct representatives v_α can be chosen effectively by finite enumeration.

For a translated cell, subtract $v_{\alpha,i}$ and $v_{\alpha,j}$ from two chosen coordinates. The same formulas recover ξ and a , so every two-coordinate projection of $\mathcal{D}_{\alpha,m}$ is injective. Distinct indices occupy distinct cosets and therefore give disjoint cells. Conversely, a tuple in one of the cells determines its coset, hence α , after which any two coordinates recover its unique parameters. This proves part (iii).

For each $\xi \in S_{\alpha,m}$ there are q choices of a , so $|\mathcal{D}_{\alpha,m}| = q|S_{\alpha,m}|$ and the remaining formula in part (ii) follows. Finally, diagonal translation by $t \in \Omega$ replaces a with $a + t$ and preserves every cell. Its action on each coordinate is the regular action of $(\Omega, +)$ on itself, proving part (iv). $\square$

The restriction $\theta_\alpha \in [0, 1]$ is harmless. Any fixed nonnegative rational multiplier c is a sum of $\lceil c \rceil$ rational numbers in $[0, 1]$, which may be assigned separate indices. The number of indices depends on c but not on N . Likewise, several independent rational factors may be assigned separate components of Ω , with their parameter subsets combined by a product.

The construction is also stable under later enlargement of Ω_{aux} . Such an enlargement preserves every ratio $|S_{\alpha,m}|/|\Omega|$, the identity $|\mathcal{D}_{\alpha,m}|/q^4 = \theta_\alpha m/(q^2 N)$ with the enlarged value of q , and pair injectivity. Consequently, all denominator-independent estimates in [Lemma 4.5](#) and [Section 5.2](#) remain valid, allowing the label set to be chosen after the other construction parameters.

4.2 Root-label cells and the finite label set

We first index the possible local declarations. Let $\mathcal{Q}_I$ be the set of unordered pairs of sorts, allowing the two sorts to coincide, and put

$$\mathcal{D}_{\mathfrak{M}} := (\mathcal{R} \times \{0, 1\}) \dot{\cup} ((\mathcal{Q}_I \setminus \mathcal{R}) \times \{*\}).$$

Thus a visible sort pair has states 0 and 1, while an invisible pair has a single dummy state. A dummy state carries no relational information; it is introduced so that the later repair procedure can give every data pair one compatible state without treating invisible pairs separately. Visible states determine the sampled pattern, whereas dummy states are ignored. In particular, if $h_{\text{st}} := |\mathcal{D}_{\mathfrak{M}}|$, then $h_{\text{st}} = |\mathcal{Q}_I| + |\mathcal{R}|$. Define the template index set

$$\mathbf{C} := \{\mathbf{v}, \text{cor}\} \dot{\cup} \{\text{ty}(i) : i \in I\} \dot{\cup} \{\text{st}(\delta) : \delta \in \mathcal{D}_{\mathfrak{M}}\}.$$

The symbols $\mathbf{v}$ and cor index the variable and correction cells; the remaining symbols index type and state cells. All pair-section cell families below are indexed by $\mathbf{C}$.

The main templates use d data vertices and $k := r - d$ root vertices. Put $P := 2r + 1$ and use $\mathbb{Z}_P$ to index the root macroblocks. Since $r \geq 4(d + 2)$, we have $k \geq \max\{5, 2d\}$. For $s \in \{k, r - 1, r - 2, k - 1\}$, let $E_s := \{0, 1, \dots, s - 1\} \subseteq \mathbb{Z}_P$. For $\mathbf{w} = (w_B)_{B \in \mathbb{Z}_P}$, set

$$R_s(\mathbf{w}) := \sum_{x \in \mathbb{Z}_P} \prod_{B \in E_s + x} w_B.$$

The *designated positions* of a translated interval are its first four positions in cyclic order. We use the abbreviations $R_{\mathbf{m}} := R_k$, $R_{\text{ty}} := R_{r-1}$, $R_{\text{st}} := R_{r-2}$, and $R_{\text{cor}} := R_{k-1}$ for the main, type, state, and correction profiles. When $d = 2$, the main and state profiles have the same support and designated positions; this is the only profile overlap that later requires special treatment.

Four rational parameters control the densities of the cell families. Introduce $\vartheta_{\text{ty}}, \vartheta_{\text{st}}, \vartheta_{\mathbf{v}}, \vartheta_{\text{cor}}$ and q_0 , subject to

$$\vartheta_{\text{ty}} = 1, \quad \vartheta_{\text{st}}, \vartheta_{\mathbf{v}}, \vartheta_{\text{cor}} \in \mathbb{Q} \cap (0, 1], \quad q_0 \in \mathbb{N}, \quad q_0 \geq 2.$$

Set $\theta_v := \vartheta_v$, $\theta_{\text{cor}} := \vartheta_{\text{cor}}$, $\theta_{\text{ty}(i)} := \vartheta_{\text{ty}}$ for $i \in I$, and $\theta_{\text{st}(\delta)} := \vartheta_{\text{st}}$ for $\delta \in \mathcal{D}_{\mathfrak{M}}$. Until Lemma 4.5, all constructions and identities are understood to depend on these provisional parameters.

To formulate the templates before the parameters are fixed permanently, apply Lemma 4.1 provisionally with index set $\mathbb{C}$, common denominator N , and lower bound q_0 . Let Ω be the resulting label set and put $q := |\Omega|$. All identities and estimates up to Lemma 4.5 are uniform over these provisional choices. After that lemma, we rebuild and fix the palette using the final parameters.

For each $B \in \mathbb{Z}_P$, let $W_B := \{W_{B,a} : a \in \Omega\}$ be a *root macroblock*; its elements are the *root labels*. Set $M := \bigcup_{B \in \mathbb{Z}_P} W_B$. Given root weights $\mathbf{u}^B = (u_a^B)_{a \in \Omega}$, write $w_B := \sum_{a \in \Omega} u_a^B$ for the total weight of macroblock B .

We distinguish two forms of root balance. The root-label weights are *uniform within macroblocks* if $u_a^B = w_B/q$ for every $B \in \mathbb{Z}_P$ and $a \in \Omega$. The macroblock totals are *balanced* if all w_B are equal. The globally balanced point introduced below satisfies both conditions and also fixes the total data mass at d/r .

For $\alpha \in \mathbb{C}$ and $S \subseteq \Omega$, put

$$\mathcal{D}_{\alpha,S} := v_\alpha + \{(a + \lambda_1 \xi, \dots, a + \lambda_4 \xi) : a \in \Omega \text{ and } \xi \in S\}.$$

For a translated support $E_s + x$, define the *lifted cylinder* by

$$\widehat{\mathcal{D}}_{\alpha,S}^{E_s+x} := \{(a_B)_{B \in E_s+x} \in \Omega^{E_s+x} : (a_x, a_{x+1}, a_{x+2}, a_{x+3}) \in \mathcal{D}_{\alpha,S}\}.$$

When $S = S_{\alpha,m}$, abbreviate this cylinder by $\widehat{\mathcal{D}}_{\alpha,m}^{E_s+x}$. Its weighted mass is

$$\mathcal{T}_{\alpha,S}^{E_s+x}(\mathbf{u}) := \sum_{(a_B)_{B \in E_s+x} \in \widehat{\mathcal{D}}_{\alpha,S}^{E_s+x}} \prod_{B \in E_s+x} u_{a_B}^B.$$

Let $\mathcal{T}_{\alpha,S}^{(s)}$ be the sum of these masses over all translates, and for the nested parameter sets from Lemma 4.1 write $\mathcal{T}_{\alpha,m}^{(s)} := \mathcal{T}_{\alpha,S_{\alpha,m}}^{(s)}$. If the weights are uniform within each macroblock, so that $u_a^B = w_B/q$, then part (ii) of that lemma gives

$$\mathcal{T}_{\alpha,m}^{(s)} = \frac{\theta_\alpha}{q^2} \frac{m}{N} R_s(\mathbf{w}). \quad (7)$$

Indeed, the undesignated coordinates are unrestricted and every permitted root tuple has the same weight.

For $\xi \in S_{v,N}$, put

$$\mathcal{D}_{v,\xi} := v_v + \{(a + \lambda_1 \xi, \dots, a + \lambda_4 \xi) : a \in \Omega\},$$

and define $\widehat{\mathcal{D}}_{v,\xi}^{E_k+x}$ by the same cylinder construction. Set

$$\mathcal{V}_\xi^{(k)}(\mathbf{u}) := \sum_{x \in \mathbb{Z}_P} \sum_{(a_B)_{B \in E_k+x} \in \widehat{\mathcal{D}}_{v,\xi}^{E_k+x}} \prod_{B \in E_k+x} u_{a_B}^B.$$

Here both the diagonal parameter a and the cyclic translate x vary. For $\Xi \subseteq S_{v,N}$, set $\mathcal{V}_\Xi^{(k)} := \sum_{\xi \in \Xi} \mathcal{V}_\xi^{(k)}$. By definition, $\mathcal{V}_\Xi^{(k)} = \mathcal{T}_{v,\Xi}^{(k)}$, and when the root-label weights are uniform within each macroblock, $\mathcal{V}_\Xi^{(k)} = |\Xi| R_m(\mathbf{w})/q^3$. These individual slices will later encode the admissible parameter set associated with a sampled pattern.

Finally, define the *residual squarefree polynomial* by

$$\mathcal{C}_{\text{res}}(\mathbf{u}) := e_k(\mathbf{w}) - \mathcal{T}_{v,N}^{(k)}(\mathbf{u}) - \mathbf{1}_{\{d=2\}} \sum_{\delta \in \mathcal{D}_{\mathfrak{M}}} \mathcal{T}_{\text{st}(\delta),N}^{(k)}(\mathbf{u}). \quad (8)$$

Thus we remove the variable cells from the complete squarefree root polynomial and, when $d = 2$, also remove the state cells that share the main support. These cells are pairwise disjoint, so

$$\mathcal{C}_{\text{res}}(\mathbf{u}) \geq 0 \quad \text{coefficientwise.} \quad (9)$$

4.3 Partial rootings and edge templates

We now turn the root-label cells into local edge rules. A partial rooting assigns some of the abstract labels in M to vertices of a finite r -graph; the six template families then specify which r -sets are available as edges.

The six families have distinct algebraic and semantic roles. The following table is a guide to the formal definition.

Family	Data degree	Root profile	Purpose
Residual aggregate	d	k , squarefree	Supplies the nonnegative squarefree baseline.
Variable	d	k , squarefree	Encodes a_β through the permitted parameters ξ .
Type	1	$r - 1$, squarefree	Assigns a sort to one data vertex.
State	2	$r - 2$, squarefree	Records a visible 0/1 relation or an invisible dummy state.
Correction	$d + 1$	$k - 1$, squarefree	Makes the common balanced point stationary.
Balancing	d	k , one repeated block	Enforces root-label uniformity and stabilizes the macroblock totals.

A *partial M -rooting* of an r -graph G is an injective map $\mu: \text{dom } \mu \rightarrow V(G)$ with $\text{dom } \mu \subseteq M$. Its image consists of the *rooted vertices*, and $U_\mu := V(G) \setminus \text{im } \mu$ is the *unrooted vertex set*. For $S \subseteq \text{dom } \mu$, write $\mu(S) := \{\mu(s) : s \in S\}$.

A partial rooting ν *extends* μ , written $\nu \supseteq \mu$, if $\text{dom } \mu \subseteq \text{dom } \nu$ and $\nu|_{\text{dom } \mu} = \mu$. Two partially M -rooted r -graphs (G, μ) and (H, ν) with the same rooting domain are *rooted-isomorphic* if some r -graph isomorphism $\varphi: G \rightarrow H$ satisfies $\varphi \circ \mu = \nu$.

For $S \subseteq \mathbb{Z}_P$ and $\mathbf{a} = (a_B)_{B \in S} \in \Omega^S$, put $W(S, \mathbf{a}) := \{W_{B, a_B} : B \in S\}$. Relative to a partial rooting μ , the tuple $\mathbf{a}$ is μ -*available* if $W(S, \mathbf{a}) \subseteq \text{dom } \mu$.

Definition 4.2. Fix a partially M -rooted r -graph (G, μ) . The elements of U_μ are the *data vertices*. The *edge templates* form the following six families of r -sets. In each family, every root-label tuple is required to be μ -available.

- (i) The *residual aggregate templates* are the r -sets $X \cup \mu(W(S, \mathbf{a}))$, where X is a d -subset of U_μ , S is a k -subset of $\mathbb{Z}_P$, and $\mathbf{a} \in \Omega^S$. If $S = E_k + x$, then $\mathbf{a} \notin \widehat{\mathcal{D}}_{v, N}^{E_k + x}$. In this case, if $d = 2$, then $\mathbf{a}$ must also lie in none of the sets $\widehat{\mathcal{D}}_{\text{st}(\delta), N}^{E_k + x}$ with $\delta \in \mathcal{D}_{\mathfrak{M}}$. If S is not a translate of E_k , then no further restriction is imposed.
- (ii) The *variable templates* are the r -sets $X \cup \mu(W(E_k + x, \mathbf{a}))$, where X is a d -subset of U_μ , $x \in \mathbb{Z}_P$, and $\mathbf{a} \in \widehat{\mathcal{D}}_{v, N}^{E_k + x}$. The *template parameter* is the unique element $\xi \in S_{v, N}$ determined by the designated coordinates of $\mathbf{a}$.
- (iii) The *type templates* are the r -sets $\{z\} \cup \mu(W(E_{r-1} + x, \mathbf{a}))$, where $i \in I$, $z \in U_\mu$, $x \in \mathbb{Z}_P$, and $\mathbf{a} \in \widehat{\mathcal{D}}_{\text{ty}(i), N}^{E_{r-1} + x}$.
- (iv) The *state templates* are the r -sets $Z \cup \mu(W(E_{r-2} + x, \mathbf{a}))$, where Z is a 2-subset of U_μ , $\delta \in \mathcal{D}_{\mathfrak{M}}$, $x \in \mathbb{Z}_P$, and $\mathbf{a} \in \widehat{\mathcal{D}}_{\text{st}(\delta), N}^{E_{r-2} + x}$.
- (v) The *correction templates* are the r -sets $Y \cup \mu(W(E_{k-1} + x, \mathbf{a}))$, where Y is a $(d + 1)$ -subset of U_μ , $x \in \mathbb{Z}_P$, and $\mathbf{a} \in \widehat{\mathcal{D}}_{\text{cor}, N}^{E_{k-1} + x}$.

(vi) The *balancing templates* are the r -sets

$$X \cup \mu(\{W_{B,a}, W_{B,b}\} \cup \{W_{C,a_C} : C \in F\}),$$

where X is a d -subset of U_μ , $B \in \mathbb{Z}_P$, $\{a, b\}$ is a 2-subset of Ω , $F \in \binom{\mathbb{Z}_P \setminus \{B\}}{k-2}$, and $(a_C)_{C \in F} \in \Omega^F$. Every root label in the display belongs to $\text{dom } \mu$.

An *available template edge* is an r -set in one of these six families. A *template representation* is a choice of one of the six template families together with all indices appearing in the corresponding item. An edge of G is *available* if it is an available template edge.

The *macro-signature* of an edge records its number of data vertices and the number of its roots in each macroblock. These signatures distinguish all nonbalancing template families except that, when $d = 2$, the main and state signatures coincide. To separate that case, define for every $x \in \mathbb{Z}_P$ the *reserved set*

$$\mathcal{R}_x := \widehat{\mathcal{D}}_{v,N}^{E_k+x} \dot{\cup} \bigcup_{\delta \in \mathcal{D}_{\mathfrak{M}}} \widehat{\mathcal{D}}_{\text{st}(\delta),N}^{E_k+x}. \quad (10)$$

The residual relation omits $\mathcal{R}_x$ on the common support, while the variable and state families restore their respective cells. Consequently,

$$(\Omega^{E_k+x} \setminus \mathcal{R}_x) \dot{\cup} \mathcal{R}_x = \Omega^{E_k+x}. \quad (11)$$

After restricting to μ -available tuples and adjoining a fixed pair of data vertices, this partition shows that the residual, variable, and state templates are disjoint and together recover the complete squarefree root polynomial on that support.

We record the resulting uniqueness of template representations, which will allow each later edge insertion or deletion to be tracked unambiguously.

Lemma 4.3. *Each available edge has exactly one representation among the six template families. The edge determines its family and all associated parameters; for a cyclic template, it also determines the translated support.*

Proof. Only a balancing-template edge uses two labels from one macroblock. Its repeated block and remaining occupied blocks therefore determine B and F , while every other template has squarefree root occupancy.

Among the squarefree templates, the number of data vertices distinguishes the main, type, state, and correction profiles, except for the main–state overlap when $d = 2$. On a translated main support, the variable cells are disjoint from the residual relation; in the exceptional case $d = 2$, the reserve decomposition (10) also separates every state cell from both. The cells with different indices lie in distinct pair-section cosets.

It remains to recover the cyclic and cell parameters. Since $P > 2r$, every interval E_s has trivial translation stabilizer, so its occupied macroblocks determine the translate. The pair-section coset then determines the index, and the designated coordinates determine the diagonal parameter a and encoded parameter ξ . Thus every representation is unique. $\square$

The finite syntax of the construction is now in place. Semantically, the type and state templates will describe an $\mathfrak{M}$ -graph on the data vertices, and the variable templates will encode the coefficient of its sampled pattern. We postpone the formal consistency conditions until the parameters have been chosen, because the inequalities required by the later repair procedure determine the relative sizes of the template families. We first analyze the balancing mechanism that supplies those inequalities.

4.4 The balancing polynomial and its stability

The balancing templates are designed to force equal weights among the root macroblocks. Continue with the cyclic-support and weight notation from [Section 4.2](#), and define the *balancing polynomial*

$$\mathcal{A}(\mathbf{u}) := e_k(\mathbf{w}) + \sum_{B \in \mathbb{Z}_P} e_2(\mathbf{u}^B) e_{k-2}(\mathbf{w}_{\neq B}). \quad (12)$$

The first term sums over squarefree root supports. The second is contributed by the balancing templates: it sums over supports with two distinct labels in one macroblock and one label in each of $k-2$ other macroblocks.

To pass from individual root labels to macroblock totals, define $J_k(\mathbf{w}) := \sum_B w_B^2 e_{k-2}(\mathbf{w}_{\neq B})$. For $0 \leq \theta \leq 1/2$, set

$$A_\theta(\mathbf{w}) := e_k(\mathbf{w}) + \theta J_k(\mathbf{w}). \quad (13)$$

Set $\theta_q := (q-1)/(2q) \in [1/4, 1/2)$. If the weights are uniform within each macroblock, then $e_2(w_B/q, \dots, w_B/q) = \theta_q w_B^2$, and hence $\mathcal{A}(\mathbf{u}) = A_{\theta_q}(\mathbf{w})$.

We first examine how A_θ changes when two macroblock weights are averaged. Write $\mathbf{w} = (x, y, \mathbf{z})$ and set $s := x + y$. Direct expansion gives

$$A_\theta(x, y, \mathbf{z}) = C_\theta(s, \mathbf{z}) + xy \Gamma_\theta(s, \mathbf{z}), \quad (14)$$

where $C_\theta(s, \mathbf{z})$ denotes the sum of the terms independent of the product xy . Define the coefficient of xy by

$$\Gamma_\theta(s, \mathbf{z}) := (1 - 2\theta) e_{k-2}(\mathbf{z}) + \theta s e_{k-3}(\mathbf{z}) + \theta \sum_i z_i^2 e_{k-4}(\mathbf{z}_{\neq i}). \quad (15)$$

Here $e_0 := 1$ and $e_j := 0$ outside its natural range. Every term in (15) is nonnegative for $0 \leq \theta \leq 1/2$. Since averaging maximizes xy with $x + y$ fixed, it cannot decrease A_θ .

Let

$$\Delta_{\text{mac}} := \{(\rho, \mathbf{w}) \geq 0 : \rho + \sum_B w_B = 1\} \quad \text{and} \quad T\Delta_{\text{mac}} := \{(a, \mathbf{z}) : a + \sum_B z_B = 0\},$$

and define

$$\Phi_\theta(\rho, \mathbf{w}) := \frac{\rho^d}{d!} A_\theta(\mathbf{w}). \quad (16)$$

Set

$$\rho_0 := \frac{d}{r}, \quad u_0 := \frac{k}{rP}, \quad \mathbf{w}_0 := u_0 \mathbf{1}, \quad \text{and} \quad p := (\rho_0, \mathbf{w}_0). \quad (17)$$

For every $\theta \in [1/4, 1/2]$, the point p maximizes Φ_θ . Indeed, repeated pair averaging equalizes the macroblock totals without decreasing the polynomial. Once they are equal, homogeneity reduces the remaining optimization to that of $\rho^d(1-\rho)^k$, whose maximum occurs at $\rho = d/r$. Put $L_\theta := \Phi_\theta(p)$. Direct substitution gives

$$L_\theta = \frac{1}{d!} \left(\frac{d}{r}\right)^d \binom{P}{k} \left(\frac{k}{rP}\right)^k \left(1 + \theta \frac{k(k-1)}{P-k+1}\right). \quad (18)$$

Write $L_- := L_{1/4}$ and $L_+ := L_{1/2}$. Put $a := k(k-1)/(P-k+1)$. Since $a \geq 0$, (18) gives $L_+/L_- = (1+a/2)/(1+a/4) \leq 2$.

We need a quantitative form of this averaging argument. The next lemma controls both the deviation of the total data weight from d/r and the deviation of the root macroblock weights from equality.

Lemma 4.4. *If $k \geq \max\{5, 2d\}$, then for every $\theta \in [1/4, 1/2]$ and every $x \in \Delta_{\text{mac}}$,*

$$\Phi_\theta(p) - \Phi_\theta(x) \geq \frac{L_\theta}{4 \cdot 6^{dr^2}} \|x - p\|_2^2. \quad (19)$$

Proof. Write $x = (\rho, \mathbf{w})$. The proof has two parts: we first control imbalance among the coordinates of $\mathbf{w}$, and then control the total mass split between ρ and $\mathbf{w}$.

For the first part, let $\mathbf{z} \in \mathbb{R}_{\geq 0}^n$ and set $Z := \sum_i z_i$. If $2 \leq m \leq n$, expansion followed by Maclaurin's inequalities gives

$$\sum_i z_i^2 e_{m-2}(\mathbf{z}_{\neq i}) = Z e_{m-1}(\mathbf{z}) - m e_m(\mathbf{z}) \geq \frac{m(m-1)}{n-m+1} e_m(\mathbf{z}). \quad (20)$$

Indeed, when $e_m > 0$, Maclaurin's inequality gives

$$\frac{e_{m-1}}{e_m} \geq \frac{m}{n-m+1} \left(\frac{\binom{n}{m}}{e_m} \right)^{1/m} \geq \frac{mn}{(n-m+1)Z}.$$

The case $e_m = 0$ is immediate. Now take $n = P-2$ and $m = k-2$. Since $(k-2)(k-3)/(P-k+1) \geq (k-2)(k-3)/(2k+2) \geq 1/2$, applying (20) to the last term of (15) and discarding the nonnegative middle term gives

$$\Gamma_\theta(s, \mathbf{z}) \geq \left(1 - 2\theta + \frac{\theta}{2}\right) e_{k-2}(\mathbf{z}) \geq \frac{1}{4} e_{k-2}(\mathbf{z}).$$

For distinct indices i and j , let $T_{ij}\mathbf{w}$ be obtained by replacing w_i and w_j with their average. The preceding bound gives

$$A_\theta(T_{ij}\mathbf{w}) - A_\theta(\mathbf{w}) \geq \frac{1}{4} (e_k(T_{ij}\mathbf{w}) - e_k(\mathbf{w})). \quad (21)$$

Put $s := \sum_B w_B$, $\bar{\mathbf{w}} := s\mathbf{1}/P$, and $V := \|\mathbf{w} - \bar{\mathbf{w}}\|_2^2$. If $s = 0$, then $\Phi_\theta(\rho, \mathbf{w}) = 0$ and $\|x - p\|_2^2 \leq 2$, so the result follows. Assume $s > 0$.

At each step, average a largest coordinate with a smallest one. If R is their difference, then $V \leq PR^2$, while the averaging reduces V by $R^2/2$ and hence by at least $V/(2P)$. The resulting sequence converges to $\bar{\mathbf{w}}$. Summing (21) along this sequence gives $A_\theta(\bar{\mathbf{w}}) - A_\theta(\mathbf{w}) \geq (e_k(\bar{\mathbf{w}}) - e_k(\mathbf{w}))/4$. Moreover, $e_2(\mathbf{w})/\binom{P}{2} = (s/P)^2(1 - PV/[(P-1)s^2])$ together with Maclaurin's inequality between e_2 and e_k , yields

$$e_k(\mathbf{w}) \leq e_k(\bar{\mathbf{w}}) \left(1 - \frac{P}{P-1} \frac{V}{s^2}\right)^{k/2}.$$

At the uniform vector, $A_\theta(\bar{\mathbf{w}})/e_k(\bar{\mathbf{w}}) = 1 + \theta k(k-1)/(P-k+1) \leq k$, because $P-k+1 > k$ and $\theta \leq 1/2$. Since $k \geq 2$, it follows that

$$A_\theta(\bar{\mathbf{w}}) - A_\theta(\mathbf{w}) \geq \frac{A_\theta(\bar{\mathbf{w}})}{4k} \frac{P}{P-1} \frac{V}{s^2}. \quad (22)$$

We next control the split between the total data and root masses. Put $s_0 := k/r$ and $\delta := \rho - \rho_0$, and define $b(\rho) := (\rho/\rho_0)^d((1-\rho)/s_0)^k$, with its continuous values at the endpoints. For $\alpha, \beta \in (0, 1)$, write

$$D_{\text{KL}}(\alpha\|\beta) := \alpha \log \frac{\alpha}{\beta} + (1-\alpha) \log \frac{1-\alpha}{1-\beta}$$

for the Bernoulli relative entropy. The identity $\log b(\rho) = -r D_{\text{KL}}(\rho_0\|\rho)$ and the Bernoulli Pinsker inequality $D_{\text{KL}}(\alpha\|\beta) \geq 2(\alpha - \beta)^2$ imply $b(\rho) \leq e^{-2r\delta^2}$. For completeness, the latter inequality follows by differentiating $D_{\text{KL}}(\alpha\|\beta) - 2(\alpha - \beta)^2$ with respect to β : the derivative

is $(\beta - \alpha)(1/(\beta(1 - \beta)) - 4)$, so the difference is minimized at $\beta = \alpha$, where it vanishes. The endpoint cases follow by continuity.

The function $y \mapsto 1 - e^{-2ry}$ is concave on $[0, 1]$, so $1 - e^{-2ry} \geq (1 - e^{-2r})y \geq y/2$. Consequently, $1 - b(\rho) \geq \delta^2/2$. If $|\delta| \geq \rho_0/2$, pair averaging and $\text{diam}^2(\Delta_{\text{mac}}) \leq 2$ give

$$\Phi_\theta(p) - \Phi_\theta(\rho, \mathbf{w}) \geq L_\theta(1 - b(\rho)) \geq \frac{d^2 L_\theta}{8r^2}.$$

This is stronger than the claimed bound.

Now suppose that $|\delta| < \rho_0/2$. If $\delta \leq 0$, then $b(\rho) \geq 2^{-d}$. If $\delta > 0$, then $\delta/s_0 \leq d/(2k) \leq 1/4$, and

$$\left(1 - \frac{\delta}{s_0}\right)^k \geq e^{-2k\delta/s_0} \geq e^{-d} > 3^{-d}.$$

Thus $b(\rho) \geq 6^{-d}$ in both cases. Applying (22) with $\bar{\mathbf{w}} = (1 - \rho)\mathbf{1}/P$ and combining it with the scalar bound above gives

$$\Phi_\theta(p) - \Phi_\theta(\rho, \mathbf{w}) \geq \frac{L_\theta}{2}\delta^2 + \frac{L_\theta}{4k6^d}V.$$

Finally, $\|x - p\|_2^2 = (1 + 1/P)\delta^2 + V$, and $k \leq r$. These two inequalities give (19). $\square$

In particular, Lemma 4.4 shows that p is the unique maximizer of Φ_θ for every $\theta \in [1/4, 1/2]$.

4.5 Choice of the construction parameters

We now choose the provisional parameters from Section 4.2. Four kinds of estimates are needed later: repair inequalities for the local declarations, a concavity bound within each root fiber, a global budget for positive auxiliary terms, and a Hessian bound for the semantic perturbation. The choices will be effective and independent of N and the coefficient vector.

Begin with a small neighborhood of the balanced point. Put $\varepsilon := 1/(2^{16}drP)$, and define $\ell := (1 - \varepsilon)u_0$, $u_+ := (1 + \varepsilon)u_0$, $\rho_- := (1 - \varepsilon)\rho_0$, and $\rho_+ := (1 + \varepsilon)\rho_0$. Let

$$U := \{(\rho, \mathbf{w}) \in \Delta_{\text{mac}} : \rho_- < \rho < \rho_+, \ell < w_B < u_+ \text{ for every } B\}.$$

Define the rational constants

$$\eta := \frac{\varepsilon^2 u_0^2}{64 \cdot 6^{dr^2}} \quad \text{and} \quad \mu_{\text{mac}} := \frac{L_-}{4 \cdot 6^{dr^2}}. \quad (23)$$

Then $0 < \eta < 1/8$. Since $L_\theta \geq L_-$, Lemma 4.4 gives $\Phi_\theta(p) - \Phi_\theta(x) \geq \mu_{\text{mac}}\|x - p\|_2^2$ uniformly for $\theta \in [1/4, 1/2]$.

We next formulate the repair requirements. Put $m_T := P\ell^{r-1}$, $m_S := P\ell^{r-2}$, $M_S := Pu_+^{r-2}$, and $M_W := Pu_+^k$. With $\vartheta_{\text{ty}} = 1$, the rational parameters $\vartheta_{\text{st}}, \vartheta_v \in (0, 1]$ must satisfy

$$\vartheta_{\text{ty}}m_T > \vartheta_{\text{st}}\rho_+M_S + \vartheta_vM_W \frac{\rho_+^{d-1}}{(d-1)!} \quad \text{and} \quad (R1)$$

$$\vartheta_{\text{st}}m_S > \vartheta_vM_W \frac{\rho_+^{d-2}}{(d-2)!}. \quad (R2)$$

After choosing these two parameters, we choose the correction density so that the semantic perturbation has zero tangent gradient at the globally balanced point. Define

$$\vartheta_{\text{cor}} := \frac{(d-1)\vartheta_{\text{ty}}\rho_0R_{\text{ty}}(\mathbf{w}_0) + (d-2)\vartheta_{\text{st}}\rho_0^2R_{\text{st}}(\mathbf{w}_0)/2}{\rho_0^{d+1}R_{\text{cor}}(\mathbf{w}_0)/(d+1)!}. \quad (24)$$

We will verify that $0 < \vartheta_{\text{cor}} \leq 1$. Thus all three cell-density parameters can be chosen effectively once r is sufficiently large.

Once the label set Ω is chosen, with $|\Omega| = q$, normalize the four densities by setting $\tau := \vartheta_{\text{ty}}/q^2$, $\sigma := \vartheta_{\text{st}}/q^2$, $c_v := \vartheta_v/q^2$, and $\gamma := \vartheta_{\text{cor}}/q^2$. Equation (24) then becomes

$$\gamma \frac{\rho_0^{d+1}}{(d+1)!} R_{\text{cor}}(\mathbf{w}_0) = (d-1)\tau\rho_0 R_{\text{ty}}(\mathbf{w}_0) + (d-2)\sigma \frac{\rho_0^2}{2} R_{\text{st}}(\mathbf{w}_0). \quad (25)$$

At weights that are uniform within each macroblock, (7) and (8) give the residual contribution

$$C_{\text{res}}(\mathbf{w}) := e_k(\mathbf{w}) - c_v R_{\text{m}}(\mathbf{w}) - \mathbf{1}_{\{d=2\}} h_{\text{st}} \sigma R_{\text{st}}(\mathbf{w}). \quad (26)$$

The semantic templates will later reduce to a single scalar parameter $t \in [0, 1]$. Balancing the total data mass among the sorts in the proportions d_i/d contributes the normalization factor $\zeta_{\mathfrak{M}} := d! \prod_{i \in I} d_i^{d_i} / (d^d \nu_{\mathfrak{M}})$. Define

$$\begin{aligned} Q_t(\rho, \mathbf{w}) &:= \tau \rho R_{\text{ty}}(\mathbf{w}) + \sigma \frac{\rho^2}{2} R_{\text{st}}(\mathbf{w}) + \gamma \frac{\rho^{d+1}}{(d+1)!} R_{\text{cor}}(\mathbf{w}) \\ &\quad + \frac{\rho^d}{d!} \left[(c_v t \zeta_{\mathfrak{M}} - c_v) R_{\text{m}}(\mathbf{w}) - \mathbf{1}_{\{d=2\}} h_{\text{st}} \sigma R_{\text{st}}(\mathbf{w}) \right]. \end{aligned} \quad (27)$$

After the sort masses are balanced, the resulting scalar polynomial is

$$\hat{H}_t(\rho, \mathbf{w}) := \Phi_{\theta_q}(\rho, \mathbf{w}) + Q_t(\rho, \mathbf{w}). \quad (28)$$

It remains to bound the number of terms that can affect concavity within a root fiber. Remove the complete squarefree and balancing contributions, and decompose each remaining pair-section term according to its sign. Call each resulting term a *signed cell profile*, and call the degree of its data factor its *data degree*. Opposite signs count separately, so variable deletion and variable restoration are distinct profiles; cyclic translates and individual parameter slices do not create additional profiles.

There are $|I|$ type-restoration profiles of data degree 1, h_{st} state-restoration profiles of degree 2, one variable-deletion and one variable-restoration profile of degree d , and one correction profile of degree $d+1$. When $d=2$, there is also one state-deletion profile for each state index. Let H_j be the number of profiles of data degree j , with $H_j = 0$ for unlisted degrees. Thus $(H_1, H_2, H_3) = (|I|, 2 + 2h_{\text{st}}, 1)$ when $d=2$; when $d > 2$, we have $H_1 = |I|$, $H_2 = h_{\text{st}}$, $H_d = 2$, and $H_{d+1} = 1$.

Put $A_{\text{fib}} := 12 \sum_j H_j / j!$ and $\bar{A}_{\text{fib}}(d) := 12(d^2 + 2d + 2)$. Since $|I| \leq d$ and $h_{\text{st}} \leq d(d+1)$, we have $A_{\text{fib}} \leq \bar{A}_{\text{fib}}(d)$. Finally, define $\underline{m}_d := (d/r)^d (k/r)^k / (4d!)$ and

$$\Delta_{r,P} := P \left(|I| + 2\mathbf{1}_{\{d>2\}} h_{\text{st}} + \frac{(d+1)^{d+1}}{(d+1)!} \right) r^{-r}.$$

The next lemma makes all choices simultaneously and records precisely the estimates used later.

Lemma 4.5. *For each integer $d \geq 2$, one can effectively compute an integer $r_{\text{aff}}(d) \geq 4(d+2)$ with the following property. For every finite binary matrix sampling scheme $\mathfrak{M}$ of total sampled order d and every $r \geq r_{\text{aff}}(d)$, one can set $\vartheta_{\text{ty}} = 1$ and effectively choose rational cell-density parameters $\vartheta_{\text{st}}, \vartheta_v, \vartheta_{\text{cor}} \in (0, 1]$ and an integer $q_0 \geq 2$ so that the following assertions hold.*

- (i) *The repair inequalities (R1)–(R2) hold and $0 < \vartheta_{\text{cor}} \leq 1$.*
- (ii) *One has*

$$\underline{m}_d \binom{P-1}{k-2} \ell^{k-2} > 2A_{\text{fib}} u_+^{k-3}. \quad (29)$$

(iii) The positive auxiliary terms satisfy

$$\Delta_{r,P} \leq \eta L_- . \quad (30)$$

(iv) The following inequalities hold for every $q \geq q_0$ and $t \in [0, 1]$:

$$\widehat{H}_t(p) \geq (1 - \eta)L_{\theta_q} \quad \text{and} \quad (O1)$$

$$\sup_{x \in \Delta_{\text{mac}}} \left\| D^2 Q_t(x) |_{T\Delta_{\text{mac}}} \right\|_{\text{op}} < \mu_{\text{mac}} . \quad (O2)$$

In particular, if $r \geq r_{\text{aff}}(d)$, then $k = r - d \geq \max\{5, 2d\}$. The integer $r_{\text{aff}}(d)$ depends only on d . All parameters in the conclusion depend only on $\mathfrak{M}$, d , and r , not on N or $(m_\beta)_\beta$.

Proof. We first choose a uniformity threshold that guarantees parts (ii) and (iii), then choose the cell-density parameters for part (i), and finally choose q_0 for part (iv).

For the auxiliary-term estimate, define $C_{\text{aux}} := |I| + 2\mathbf{1}_{\{d \geq 2\}} h_{\text{st}} + (d+1)^{d+1}/(d+1)!$ and $\overline{C}_d := d + 2d(d+1) + (d+1)^{d+1}/(d+1)!$. The bounds on $|I|$ and h_{st} give $C_{\text{aux}} \leq \overline{C}_d$.

We repeatedly use the following effective tail bound. For fixed rational numbers $B > 0$, $m \geq 0$, and $\alpha \in \{1/2, 3/4\}$, one can compute R such that $Br^m \alpha^{r-d} < 1$ for every $r \geq R$. Indeed, for $r \geq 8m$, the ratio of consecutive terms is at most $\alpha(1 + 1/r)^m \leq \alpha/(1 - m/r) \leq 6/7$. Once the inequality holds, it continues to hold, so a finite search finds such an R .

Begin with $r \geq 4(d+2)$. Then $k \geq \max\{5, 2d\}$, $k \geq 3r/4$, and $P \leq 3k$. Moreover, $\binom{P}{k} \geq \binom{2k}{k} \geq 4^k/(2k+1)$, because the central binomial coefficient is the largest of the $2k+1$ coefficients whose sum is 4^k . Hence

$$L_- \geq \frac{d^d}{d!(2k+1)} \left(\frac{4}{3}\right)^k r^{-r} . \quad (31)$$

By (23), $\eta = k^2/(2^{38}6^d d^2 r^6 P^4)$. Since $k \geq 3r/4$ and $P \leq 3r$, we obtain

$$\eta \geq 2^{-46} 6^{-d} d^{-2} r^{-8} . \quad (32)$$

On the other hand, (31) gives

$$\frac{\Delta_{r,P}}{L_-} \leq \frac{\overline{C}_d d!}{d^d} P(2k+1) \left(\frac{3}{4}\right)^k .$$

Since $P(2k+1) \leq 9r^2$, the tail bound with $\alpha = 3/4$ gives $r^8 \Delta_{r,P}/L_- < 2^{-46} 6^{-d} d^{-2}$ for all sufficiently large r , with an effective threshold. Together with (32), this proves the global budget (30).

For (29), note that $\varepsilon \leq 1/2$ and $k/r \geq 3/4$. Hence $\ell \geq 1/(4P)$. Bernoulli's inequality gives

$$\left(\frac{\ell}{u_+}\right)^{k-3} = \left(\frac{1-\varepsilon}{1+\varepsilon}\right)^{k-3} \geq 1 - 2(k-3)\varepsilon \geq \frac{1}{2} .$$

Since $P-1 = 2r \geq 2(k-2)$, we have $\binom{P-1}{k-2} \geq \binom{2(k-2)}{k-2} \geq 2^{k-2}$. It follows that $\binom{P-1}{k-2} \ell^{k-2}/u_+^{k-3} \geq 2^{k-5}/P$. Also,

$$\underline{m}_d^{-1} = 4d! \left(\frac{r}{d}\right)^d \left(1 + \frac{d}{k}\right)^k \leq 4d! 3^d \left(\frac{r}{d}\right)^d .$$

Using $P \leq 3r$, we obtain

$$\underline{m}_d \frac{\binom{P-1}{k-2} \ell^{k-2}}{u_+^{k-3}} \geq \frac{d^d 2^{k-5}}{12d! 3^d r^{d+1}} .$$

The tail bound with $\alpha = 1/2$ shows that this quantity eventually exceeds $2\bar{A}_{\text{fib}}(d)$ and hence $2A_{\text{fib}}$, with an effective threshold depending only on d . This proves the fiber-domination inequality (29).

We now choose the cell densities. First choose a positive dyadic $\vartheta_{\text{st}} \leq 1$ such that $\vartheta_{\text{st}}\rho_+M_S < m_T/2$. Then choose a positive dyadic $\vartheta_v \leq 1$ so that

$$\vartheta_v M_W \frac{\rho_+^{d-1}}{(d-1)!} < m_T - \vartheta_{\text{st}}\rho_+M_S \quad \text{and} \quad \vartheta_v M_W \frac{\rho_+^{d-2}}{(d-2)!} < \vartheta_{\text{st}}m_S.$$

These choices give both strict repair inequalities. Since $R_s(\mathbf{w}_0) = Pu_0^s$, $\rho_0 = d/r$, and $u_0 = k/(rP)$, formula (24) simplifies to

$$\vartheta_{\text{cor}} = (d+1)! \left[(d-1) \left(\frac{k}{dP} \right)^d + \frac{d-2}{2} \vartheta_{\text{st}} \left(\frac{k}{dP} \right)^{d-1} \right].$$

Put $c_d := (d+1)!(d^2 - d - 1)/(2d)^d$. Because $k/P < 1/2$ and $0 < \vartheta_{\text{st}} \leq 1$, we have $0 < \vartheta_{\text{cor}} \leq c_d$. Also $c_2 = 3/8$ and $c_3 = 5/9$. For $d \geq 3$, the first two factors in

$$\frac{c_{d+1}}{c_d} = \frac{d+2}{2(d+1)} \frac{d^2 + d - 1}{d^2 - d - 1} \left(\frac{d}{d+1} \right)^d$$

are at most $5/8$ and $11/5$, respectively. The binomial theorem gives $(1+1/d)^d \geq 2+(d-1)/(2d) \geq 7/3$, so the last factor is at most $3/7$. Consequently, $c_{d+1}/c_d \leq 33/56 < 1$. Thus $c_d < 1$ for every $d \geq 2$, and $\vartheta_{\text{cor}} \in (0, 1]$ without any further increase in the threshold. This proves part (i).

It remains to choose q_0 . Define $\tilde{Q}_t := q^2 Q_t$. This is a fixed degree- r polynomial whose coefficients are affine in t and independent of q , N , and the coefficient vector. Expand it in the monomial basis, and let S_* be one plus the sum of the absolute values of all constant and t -coefficients. On the simplex, the sum of the absolute values of the second partial derivatives of any monomial of degree at most r is at most $r(r-1)$. Since the operator norm of a Hessian is at most its entrywise ℓ^1 -norm, $\|D^2 \tilde{Q}_t|_{T\Delta_{\text{mac}}}\|_{\text{op}} \leq r(r-1)S_*$; also $|\tilde{Q}_t(p)| \leq S_*$. Choose q_0 so that

$$q_0^2 > \max \left\{ \frac{2S_*}{\eta L_-}, \frac{4 \cdot 6^d r^3 (r-1) S_*}{L_-} \right\}.$$

The first comparison gives (O1); the second, together with (23), gives (O2). This proves part (iv). Every threshold used above comes from a finite comparison of rational numbers, so $r_{\text{aff}}(d)$ and all stated parameters are computable. $\square$

From now on, assume $r \geq r_{\text{aff}}(d)$ and fix the cell-density parameters and q_0 supplied by Lemma 4.5. Apply Lemma 4.1 with these choices to fix Ω and all associated cells. Thus every preceding definition is now permanent.

4.6 Local declarations and admissibility

The templates provide the finite syntax of the construction; we now specify their semantics. By Lemma 4.3, every available edge uniquely determines its family and parameters. Type, state, and variable edges therefore make unambiguous local declarations on their data vertices.

Fix a partial rooting μ of an r -graph G . A vertex $z \in U_\mu$ carries type i if some edge of G has a type-template representation with data set $\{z\}$ and index $\text{ty}(i)$. Let Ty_μ^i be the set of all such vertices. Similarly, a pair $Z \in \binom{U_\mu}{2}$ carries state δ if some edge has a state-template

representation with data set Z and index $\text{st}(\delta)$; let St_μ^δ be the set of all such pairs. Finally, for $X \in \binom{U_\mu}{d}$, define its set of declared variable parameters by

$$\Xi_\mu(X) := \left\{ \xi \in S_{v,N} : \begin{array}{l} \text{some edge of } G \text{ has a variable-template representation} \\ \text{with data set } X \text{ and parameter } \xi \end{array} \right\}.$$

These types, states, and parameters are the *local declarations* of (G, μ) .

A data set is *fully typed* if each of its vertices carries exactly one type. Such a set has *composition* $\mathbf{d}$ if it contains exactly d_i vertices of type i for every $i \in I$. If a fully typed d -set X has composition $\mathbf{d}$ and every visible pair in X carries exactly one state compatible with its endpoint types, these states determine a pattern $\beta_X \in \mathcal{U}_M$.

A partially M -rooted r -graph (G, μ) is *admissible* if every edge of G is available in the sense of [Definition 4.2](#) and the following five consistency conditions hold.

- (A1) The sets Ty_μ^i , $i \in I$, are pairwise disjoint.
- (A2) The relations St_μ^δ , $\delta \in \mathcal{D}_M$, are pairwise disjoint.
- (A3) If $Z \in \text{St}_\mu^\delta$ and both endpoints of Z carry exactly one type, then the sort pair underlying δ must agree with the types of the two endpoints.
- (A4) If $\Xi_\mu(X) \neq \emptyset$ and X is fully typed, then X has composition $\mathbf{d}$.
- (A5) If $\xi \in \Xi_\mu(X)$ and β_X is defined, then $\xi \in S_{v, m_{\beta_X}}$.

Conditions (A1) and (A2) enforce uniqueness, condition (A3) enforces compatibility, and conditions (A4) and (A5) encode the sampled pattern and its coefficient. The last condition is deliberately one-sided: it requires $\Xi_\mu(X) \subseteq S_{v, m_{\beta_X}}$, rather than equality. Equality would not be preserved when edges are deleted and would therefore destroy monotonicity; moreover, under a partial declaration the pattern β_X need not yet be defined. The repair procedure first completes the types and states, deletes incompatible variable orbits, and then inserts every remaining permitted parameter. Thus equality is recovered at the optimizing stage. Since $|S_{v, m_\beta}|/q = \vartheta_v a_\beta$, the resulting variable contribution is proportional to the input coefficient a_β .

None of the five conditions requires a declaration to be present; they constrain only declarations that occur. Since all relevant sets are finite, admissibility is decidable uniformly from (G, μ) and the construction data.

Definition 4.6. The *admissible class* $\mathcal{C}_f$ consists of the finite r -graphs G for which (G, μ) is admissible for some partial M -rooting μ .

4.7 Bounded witnesses and finite obstructions

The admissibility predicate is local but uses a partial rooting. To obtain an unrooted obstruction family, we use a general search-tree criterion: every failed rooting has a bounded witness, and any admissible extension must place a new root inside that witness. Since only finitely many labels are available, iterating this condition produces a bounded unrooted obstruction.

Fix an integer $r \geq 2$, a label set M of size s , and a rooted-isomorphism-invariant predicate Adm on finite partially M -rooted r -graphs. Call (G, μ) *Adm-admissible* when $\text{Adm}(G, \mu)$ holds, and let

$$\mathcal{D} := \{G : G \text{ admits an Adm-admissible partial } M\text{-rooting}\}.$$

For $w \geq 1$, say that Adm has the *w-bounded-witness property* if every inadmissible (G, μ) has a subgraph $W_\mu \subseteq G$ with $v(W_\mu) \leq w$ such that the following holds: whenever $W_\mu \subseteq H \subseteq G$ and (H, ν) is Adm-admissible with $\nu \supseteq \mu$, some newly rooted label is mapped to a vertex of W_μ that was unrooted under μ . Equivalently,

$$\nu(\text{dom } \nu \setminus \text{dom } \mu) \cap (V(W_\mu) \setminus \text{im } \mu) \neq \emptyset.$$

The following search-tree argument converts this bounded extension condition into a finite unrooted obstruction family.

Proposition 4.7. *If $w \geq 1$, the class $\mathcal{D}$ is monotone, and Adm has the w -bounded-witness property, then $\mathcal{D} = \text{Forb}(\mathcal{B})$ for some finite family $\mathcal{B}$ of r -graphs, and every $B \in \mathcal{B}$ satisfies $v(B) \leq w \sum_{j=0}^s (sw)^j$. If, in addition, r , M , and w are given effectively and $\text{Adm}(G, \mu)$ is decidable uniformly for finite pairs (G, μ) , then $\mathcal{B}$ is effectively computable.*

Proof. Fix $G \notin \mathcal{D}$. Build a search tree whose nodes are partial rootings of G , beginning with the empty map. At a node μ , choose a witness W_μ . For every unused label $m \in M$ and every $v \in V(W_\mu) \setminus \text{im } \mu$, add the child $\mu \cup \{m \mapsto v\}$. Since G has no admissible partial rooting, a witness exists at every node. The tree has depth at most s , branching factor at most sw , and hence at most $\sum_{j=0}^s (sw)^j$ nodes.

Let H_* be the union of all selected witness subgraphs. Induction on depth shows that the image of each node rooting lies in the witnesses along its root-to-node path and hence in $V(H_*)$. Therefore $v(H_*) \leq w \sum_{j=0}^s (sw)^j$.

We claim that $H_* \notin \mathcal{D}$. Otherwise, let η be an admissible partial rooting of H_* . Starting from $\mu_0 = \emptyset$, suppose that the current node satisfies $\mu_j \subsetneq \eta$. Because H_* contains W_{μ_j} , the bounded-witness property supplies an $m \in \text{dom } \eta \setminus \text{dom } \mu_j$ with $\eta(m) \in V(W_{\mu_j}) \setminus \text{im } \mu_j$. Hence the search tree contains the child

$$\mu_{j+1} := \mu_j \cup \{m \mapsto \eta(m)\}$$

which is again extended by η and has a strictly larger domain. After at most $|\text{dom } \eta|$ steps, this path reaches the node η . Applying the bounded-witness property once more would require a label in $\text{dom } \eta \setminus \text{dom } \eta$, a contradiction.

Let $\mathcal{B}$ contain one representative of every isomorphism class of r -graphs outside $\mathcal{D}$ with at most the asserted number of vertices. Every graph outside $\mathcal{D}$ contains a subgraph such as H_* , while monotonicity prevents a graph in $\mathcal{D}$ from containing a member of $\mathcal{B}$. Thus $\mathcal{D} = \text{Forb}(\mathcal{B})$.

Under the effective hypotheses, enumerate all r -graphs up to the stated order and all their partial injective rootings. The predicate Adm decides whether any such rooting is admissible, so membership in $\mathcal{D}$, and hence the family $\mathcal{B}$, is computable. $\square$

Proof of Lemma 3.1. We verify the two hypotheses of Proposition 4.7. First, $\mathcal{C}_f$ is monotone. Indeed, let $G \in \mathcal{C}_f$ have admissible partial rooting μ , and let $H \subseteq G$. Restrict μ to the labels whose images lie in $V(H)$. Every edge of H retains its template representation, while passing to H can only delete local declarations. By conditions (A1) and (A2), each data vertex has at most one type and each data pair at most one state. Deleting declarations therefore either makes a hypothesis in conditions (A3)–(A5) false or leaves all relevant declarations, including β_X , unchanged. Thus the restricted rooting is admissible and $H \in \mathcal{C}_f$.

We next bound the size of a witness to inadmissibility. Let

$$v_{\mathfrak{M}} := \sum_{\substack{i \in I \\ \{i, i\} \in \mathcal{R}}} \binom{d_i}{2} + \sum_{\substack{i < j \\ \{i, j\} \in \mathcal{R}}} d_i d_j.$$

A template failure is witnessed by one edge, and a failure of type or state uniqueness by two declaration edges. A state–type incompatibility uses one state edge and at most two type edges; a variable-composition failure uses one variable edge and at most d type edges. Finally, a variable-parameter failure is witnessed by one variable edge, at most d type edges, and at most $v_{\mathfrak{M}}$ visible-state edges. Thus every failure has a witness on at most $w := r(1 + d + v_{\mathfrak{M}})$ vertices.

It remains to show that these witnesses persist under rooting extensions. Order the possible failures by placing template failure first, followed by conditions (A1)–(A5). For an inadmissible (G, μ) , let W_μ witness the first failure in this order. Suppose that $W_\mu \subseteq H \subseteq G$ and that $\nu \supseteq \mu$ maps no new root label into $V(W_\mu) \setminus \text{im } \mu$. Every witness vertex unrooted under μ then remains unrooted under ν , so the template representations and declarations witnessed inside W_μ persist.

A template failure, type conflict, or state conflict therefore remains a violation. If W_μ witnesses condition (A3), its state declaration and endpoint types persist. A new, different endpoint type violates condition (A1); otherwise the same incompatible state still violates condition (A3). The argument for condition (A4) is identical: either a new type violates uniqueness, or X retains its fully typed but incorrect composition.

For condition (A5), choose W_μ to contain the variable edge witnessing $\xi \in \Xi_\mu(X)$ and the type and visible-state edges defining β_X . All these declarations persist. A new type on X either repeats the existing type or violates condition (A1). Likewise, a new state with an incompatible sort pair violates condition (A3); on a visible pair, a compatible new state either repeats the existing state or violates condition (A2). A dummy state lies on an invisible pair and does not affect β_X . Hence either an earlier admissibility condition fails or β_X remains unchanged, in which case the original nonmembership $\xi \notin S_{v, m_{\beta_X}}$ still violates condition (A5).

Thus ν cannot be admissible unless it maps a new root label into $V(W_\mu) \setminus \text{im } \mu$. The predicate has the w -bounded-witness property, and it is decidable and invariant under rooted isomorphism. Applying Proposition 4.7 with $\mathcal{D} = \mathcal{C}_f$ gives an effectively computable finite family $\mathcal{A}_f$ satisfying $\mathcal{C}_f = \text{Forb}(\mathcal{A}_f)$.

It remains only to show that the class contains an edge. The cell construction uses fewer than q^2 pair-section cosets. On one translated main support, choose a k -tuple of root labels whose designated entries lie in an unused coset, and adjoin d new data vertices. The resulting r -set is a residual edge. The r -graph consisting of this single edge admits an admissible partial M -rooting, so it belongs to $\mathcal{C}_f$.

Finally, Lemma 4.5 gives an effective threshold depending only on d , while the cell construction, template list, admissibility test, and bounded-witness enumeration are all effective and finite. This proves Lemma 3.1. $\square$

5 Proof of the affine-evaluation lemma

We prove Lemma 3.2 by reducing the entire optimization to the single scalar $t = \text{Val}_{\mathfrak{M}}(f)$. First, link completion gives an exact formula for the Lagrangian polynomial. We then combine root-fiber concavity with the global balancing estimates from Lemmas 4.4 and 4.5 to force exact balance among the root labels. A repair procedure completes the type, state, and variable declarations without decreasing the value. This turns the class Lagrangian into the maximum of a scalar polynomial H_t . Finally, all polynomials H_t , $t \in [0, 1]$, are shown to have the same maximizer, where evaluation is affine in t with positive rational slope.

5.1 Link completion and the exact Lagrangian polynomial

We begin by completing every local declaration to its full family of template edges. This makes the contribution of each template family explicit.

For a type declaration $z \in \text{Ty}_\mu^i$, the *type bundle* consists of all r -sets $\{z\} \cup \mu(W(E_{r-1} + x, \mathbf{a}))$ with $x \in \mathbb{Z}_P$ and $\mathbf{a} \in \widehat{\mathcal{D}}_{\text{ty}(i), N}^{E_{r-1}+x}$. For a state declaration $\{z, z'\} \in \text{St}_\mu^\delta$, the *state bundle* consists of all r -sets $\{z, z'\} \cup \mu(W(E_{r-2} + x, \mathbf{a}))$ with $x \in \mathbb{Z}_P$ and $\mathbf{a} \in \widehat{\mathcal{D}}_{\text{st}(\delta), N}^{E_{r-2}+x}$. A bundle is *complete* if all its members are edges of G .

Likewise, for $X \in \binom{U_d^\mu}{d}$ and $\xi \in \Xi_\mu(X)$, the *variable orbit* consists of all r -sets $X \cup \mu(W(E_k + x, \mathbf{a}))$ with $x \in \mathbb{Z}_P$ and $\mathbf{a} \in \widehat{\mathcal{D}}_{v,\xi}^{E_k+x}$; it is complete when all these sets are edges. An admissible rooting μ is *link-complete* if it is total, G contains every residual aggregate, balancing, and correction template, and every local declaration present in G has its complete bundle or orbit.

Any admissible graph can be enlarged to this completed form without changing its existing structure.

Lemma 5.1. *For every $G \in \mathcal{C}_\dagger$, there is an $H \in \mathcal{C}_\dagger$ such that $G \subseteq H$ and H admits a link-complete rooting.*

Proof. Fix an admissible partial rooting μ witnessing $G \in \mathcal{C}_\dagger$. Add one new vertex for each label in $M \setminus \text{dom } \mu$ and extend μ to a total rooting. The old data vertices remain data vertices, so every old edge keeps its template representation. Now insert all residual aggregate, balancing, and correction templates; these unconditional families create no local declarations.

Complete the bundle or orbit of every declaration already present. By Lemma 4.3, each inserted edge has exactly the same family, data set, and indices as the declaration being completed, so no new declaration appears. Thus the sets Ty_μ^i , St_μ^δ , and $\Xi_\mu(X)$ are unchanged, and conditions (A1)–(A5) remain valid. When $d = 2$, (10) also guarantees that every restored edge is disjoint from the residual relation. The resulting graph is admissible and link-complete. $\square$

Fix a link-complete rooted graph (G, μ) with probability weighting ω . For $z \in U_\mu$, write $x_z := \omega_z$, and let $E_j(\mathbf{x}) := e_j((x_z)_{z \in U_\mu})$. For a root label $W_{B,a}$, set $u_a^B := \omega_{\mu(W_{B,a})}$, $\mathbf{u}^B := (u_a^B)_{a \in \Omega}$, and $w_B := \sum_{a \in \Omega} u_a^B$.

For $i \in I$, let ρ_i^μ be the total weight of Ty_μ^i ; thus $\rho_i^\mu := \sum_{z \in \text{Ty}_\mu^i} x_z$. For $\delta \in \mathcal{D}_\mathfrak{M}$, put $\Delta_\delta^\mu := \sum_{\{z,z'\} \in \text{St}_\mu^\delta} x_z x_{z'}$. Finally, define

$$\mathcal{B}_*(\mathbf{x}, \mathbf{u}) := E_d(\mathbf{x}) \sum_{B \in \mathbb{Z}_P} e_2(\mathbf{u}^B) e_{k-2}(\mathbf{w}_{\neq B}). \quad (33)$$

This is precisely the contribution of the balancing templates.

Link completeness now lets us sum the six disjoint template families exactly.

Proposition 5.2. *For every link-complete rooted graph (G, μ) with a probability weighting ω , its Lagrangian polynomial is*

$$\begin{aligned} p_G(\omega) = & E_d(\mathbf{x}) \mathcal{C}_{\text{res}}(\mathbf{u}) + \mathcal{B}_*(\mathbf{x}, \mathbf{u}) + E_{d+1}(\mathbf{x}) \mathcal{T}_{\text{cor},N}^{(k-1)}(\mathbf{u}) \\ & + \sum_{i \in I} \rho_i^\mu \mathcal{T}_{\text{ty}(i),N}^{(r-1)}(\mathbf{u}) + \sum_{\delta \in \mathcal{D}_\mathfrak{M}} \Delta_\delta^\mu \mathcal{T}_{\text{st}(\delta),N}^{(r-2)}(\mathbf{u}) + \sum_{X \in \binom{U_d^\mu}{d}} \left(\prod_{z \in X} x_z \right) \mathcal{V}_{\Xi_\mu(X)}^{(k)}(\mathbf{u}). \end{aligned} \quad (34)$$

The six summands are the contributions of six pairwise disjoint template edge families.

Proof. The first three terms come from the unconditional template families. The residual aggregate and balancing templates contribute $E_d(\mathbf{x}) \mathcal{C}_{\text{res}}(\mathbf{u})$ and $\mathcal{B}_*(\mathbf{x}, \mathbf{u})$, respectively, while completeness of the correction templates gives $E_{d+1}(\mathbf{x}) \mathcal{T}_{\text{cor},N}^{(k-1)}(\mathbf{u})$.

The remaining three terms come from local declarations. Summing the complete type bundles first over vertices and then over i gives $\sum_i \rho_i^\mu \mathcal{T}_{\text{ty}(i),N}^{(r-1)}(\mathbf{u})$. The same argument for declared pairs gives $\sum_\delta \Delta_\delta^\mu \mathcal{T}_{\text{st}(\delta),N}^{(r-2)}(\mathbf{u})$. For each $X \in \binom{U_d^\mu}{d}$, the complete variable orbits with parameters in $\Xi_\mu(X)$ contribute $(\prod_{z \in X} x_z) \mathcal{V}_{\Xi_\mu(X)}^{(k)}(\mathbf{u})$. All edges are accounted for because every unconditional template is present and every declared bundle or orbit is complete. The contributions are disjoint by Lemma 4.3; when $d = 2$, separation of the residual, variable, and state terms also uses (10). $\square$

At weights that are uniform within each root macroblock, $u_a^B = w_B/q$, the cell-density identities turn (34) into

$$\begin{aligned} p_G(\omega) = & E_d(\mathbf{x})C_{\text{res}}(\mathbf{w}) + \theta_q E_d(\mathbf{x})J_k(\mathbf{w}) + \gamma E_{d+1}(\mathbf{x})R_{\text{cor}}(\mathbf{w}) \\ & + \tau R_{\text{ty}}(\mathbf{w}) \sum_i \rho_i^\mu + \sigma R_{\text{st}}(\mathbf{w}) \sum_\delta \Delta_\delta^\mu + \frac{R_m(\mathbf{w})}{q^3} \sum_{X \in \binom{U_d^\mu}{d}} |\Xi_\mu(X)| \prod_{z \in X} x_z. \end{aligned} \quad (35)$$

5.2 Root-fiber concavity and the global balancing bound

We now optimize the distribution of weight inside each root macroblock while keeping the data weights and macroblock totals fixed. On such a fiber, the balancing term has a uniformly negative Hessian, whereas all remaining cell terms have a denominator-independent Hessian bound. The parameter choice in Lemma 4.5 makes the first dominate the second.

Let $\mathbf{1}^\perp := \{z \in \mathbb{R}^\Omega : \sum_{a \in \Omega} z_a = 0\}$. For fixed macroblock totals $\mathbf{w}$, let

$$\mathcal{X}(\mathbf{w}) := \prod_B \left\{ \mathbf{u}^B \geq 0 : \sum_a u_a^B = w_B \right\} \quad \text{and} \quad T_{\text{fib}} := \bigoplus_B \mathbf{1}^\perp.$$

For $\mathbf{u} \in \mathcal{X}(\mathbf{w})$, let $\omega_{\mathbf{u}}$ agree with ω on the data vertices and assign weight u_a^B to the root $\mu(W_{B,a})$.

We first obtain a uniform Hessian estimate for a single pair-section term; summing this estimate will later control every nonbalancing profile.

Lemma 5.3. *Fix $\alpha \in \mathbb{C}$, $S \subseteq \Omega$, $s \in \{k, r-1, r-2, k-1\}$, and $y \in \mathbb{Z}_P$. Write $B_i := y + i - 1$ for $i \in [4]$. Let the root-weight vectors $\mathbf{u}^B$, $B \in E_s + y$, be nonnegative and set $U_B := \sum_a u_a^B$. If $i, j \in [4]$ are distinct, then*

$$\left\| \frac{\partial^2 \mathcal{T}_{\alpha, S}^{E_s + y}}{\partial \mathbf{u}^{B_i} \partial \mathbf{u}^{B_j}} \right\|_{\text{op}} \leq \prod_{B \in (E_s + y) \setminus \{B_i, B_j\}} U_B.$$

Consequently, if $A, w_+ \in \mathbb{R}_{\geq 0}$, $U_B \leq w_+$ for every $B \in E_s + y$, and $z \in T_{\text{fib}}$, then

$$\left| D^2 \left(A \mathcal{T}_{\alpha, S}^{E_s + y} \right) [z, z] \right| \leq 3A w_+^{s-2} \sum_{i=1}^4 \|z^{B_i}\|_2^2.$$

All numerical constants in these bounds are independent of q , N , S , and the cell-density parameter.

Proof. Fix distinct $i, j \in [4]$, and let M_{ij} be the corresponding mixed Hessian block. For $c \in \Omega$, let $\mathcal{C}_c$ be the set of tuples in $\widehat{\mathcal{D}}_{\alpha, S}^{E_s + y}$ whose B_i -coordinate is c , and project away the B_i - and B_j -coordinates via π_{ij} . For fixed $\xi \in S$, the equation $c = v_{\alpha, i} + a + \lambda_i \xi$ determines a . Conversely, c together with either remaining designated coordinate determines ξ by the unit-difference property in part (i) of Lemma 4.1. Hence π_{ij} is injective on $\mathcal{C}_c$, with image contained in

$$\prod_{B \in (E_s + y) \setminus \{B_i, B_j\}} \Omega.$$

All entries of M_{ij} are nonnegative, so

$$\sum_{c' \in \Omega} (M_{ij})_{c, c'} = \sum_{\mathbf{a} \in \pi_{ij}(\mathcal{C}_c)} \prod_{B \in (E_s + y) \setminus \{B_i, B_j\}} u_{a_B}^B \leq \prod_{B \in (E_s + y) \setminus \{B_i, B_j\}} U_B.$$

Interchanging i and j gives the same estimate for every column sum, so the Schur bound proves the operator-norm inequality.

For the quadratic-form estimate, observe that each undesignated vector enters $\mathcal{T}_{\alpha,S}^{E_s+y}$ only through the sum of its coordinates. Every Hessian term involving such a block therefore vanishes on T_{fib} , and all diagonal blocks vanish because the polynomial is affine in each coordinate vector. The mixed-block estimate above now gives

$$\left| D^2 \mathcal{T}_{\alpha,S}^{E_s+y}[z, z] \right| \leq 2w_+^{s-2} \sum_{1 \leq i < j \leq 4} \|z^{B_i}\|_2 \|z^{B_j}\|_2 \leq 3w_+^{s-2} \sum_{i=1}^4 \|z^{B_i}\|_2^2.$$

Multiplication by A completes the proof. $\square$

For the balancing term itself, the fiber Hessian can be computed exactly and has the required negative sign.

Lemma 5.4. *For every $z = (z^B)_B \in T_{\text{fib}}$,*

$$D^2 \mathcal{B}_*(\mathbf{x}, \mathbf{u})[z, z] = -E_d(\mathbf{x}) \sum_B e_{k-2}(\mathbf{w}_{\neq B}) \|z^B\|_2^2. \quad (36)$$

If $E_d(\mathbf{x}) \geq \underline{m}_d$ and $\ell \leq w_B \leq u_+$ for every B , then

$$D^2 \mathcal{B}_*[z, z] \leq -\underline{m}_d \binom{P-1}{k-2} \ell^{k-2} \|z\|_2^2.$$

Indeed, on a fixed-total fiber, each factor $e_{k-2}(\mathbf{w}_{\neq B})$ is constant and $D^2 e_2(\mathbf{u}^B)[z^B, z^B] = -\|z^B\|_2^2$. Moreover, each of the $\binom{P-1}{k-2}$ monomials in $e_{k-2}(\mathbf{w}_{\neq B})$ is at least ℓ^{k-2} , which gives the stated inequality.

Let $\mathcal{P}_{\text{rem}}(\mathbf{u}) := p_G(\omega_{\mathbf{u}}) - E_d(\mathbf{x})e_k(\mathbf{w}) - \mathcal{B}_*(\mathbf{x}, \mathbf{u})$ be the sum of all remaining signed cell profiles. Expanding $\mathcal{C}_{\text{res}}$ turns the deleted and restored cells into precisely these signed profiles. The pair-section estimate gives a root-fiber Hessian bound for the remainder that is independent of the denominator.

Lemma 5.5. *If $\ell \leq w_B \leq u_+$ for every B , then for every $\mathbf{u} \in \mathcal{X}(\mathbf{w})$ and every $z \in T_{\text{fib}}$,*

$$|D^2 \mathcal{P}_{\text{rem}}(\mathbf{u})[z, z]| \leq A_{\text{fib}} u_+^{k-3} \|z\|_2^2.$$

This bound is uniform in N, q , the coefficient vector, the graph, and the variable-parameter subsets.

Proof. Recall the profile enumeration preceding Lemma 4.5. The type, state-restoration, variable-deletion, variable-restoration, and correction profiles have degree pairs $(1, r-1)$, $(2, r-2)$, (d, k) , (d, k) , and $(d+1, k-1)$, with multiplicities $|I|$, h_{st} , 1, 1, and 1. When $d=2$, there are also h_{st} state-deletion profiles of degree pair $(2, k)$. Grouping by data degree gives the numbers H_j in A_{fib} .

Consider one signed cell profile of data degree j . By Lemma 5.3, a translated support contributes at most three times its data factor times the remaining root totals. Across all cyclic translates, each macroblock occurs once in each of the four designated positions, so the cyclic sum contributes the factor $3 \cdot 4 = 12$. The profile is therefore bounded by $12E_j(\mathbf{x})u_+^{r-j-2} \|z\|_2^2$.

For a fixed data d -set X , pair injectivity makes the root slices indexed by $\Xi_\mu(X)$ disjoint, and their union lies in the single cell $\mathcal{D}_{\mathbf{v}, S_{\mathbf{v}}, N}$. They can therefore be treated as one variable-restoration profile, with no factor $|\Xi_\mu(X)|$. Summing the associated data monomials over X gives $E_d(\mathbf{x}) \leq 1/d!$. Summing all signed profiles yields $|D^2 \mathcal{P}_{\text{rem}}(\mathbf{u})[z, z]| \leq 12 \sum_j (H_j/j!) u_+^{r-j-2} \|z\|_2^2$. Since $u_+ = (1+\varepsilon)k/(rP) < 2/P < 1$ and every signed profile has root degree at least $k-1$, we have $r-j-2 \geq k-3$ and hence $u_+^{r-j-2} \leq u_+^{k-3}$. The stated bound follows. $\square$

The negative Hessian from the balancing templates now dominates the remainder: combining Lemmas 5.4 and 5.5 with (29) gives the following.

Corollary 5.6. *If $E_d(\mathbf{x}) \geq \underline{m}_d$ and $\ell \leq w_B \leq u_+$ for every B , then the map $\mathbf{u} \mapsto p_G(\omega_{\mathbf{u}})$ is strictly concave on $\mathcal{X}(\mathbf{w})$. Moreover, for every $\mathbf{u} \in \mathcal{X}(\mathbf{w})$ and $z \in T_{\text{fib}}$, its Hessian satisfies*

$$D^2 p_G(\omega_{\mathbf{u}})[z, z] \leq -\frac{1}{2} \underline{m}_d \binom{P-1}{k-2} \ell^{k-2} \|z\|_2^2.$$

To use this local concavity estimate globally, we also need a bound that separates the main balancing contribution from all positive auxiliary profiles.

Lemma 5.7. *Every weighting of a link-complete admissible rooted graph satisfies*

$$p_G(\omega) \leq E_d(\mathbf{x}) A_{\theta_q}(\mathbf{w}) + \Delta_{r,P} \leq \Phi_{\theta_q}(\rho, \mathbf{w}) + \Delta_{r,P}, \quad (37)$$

where $\rho := \sum_z x_z$.

Proof. We first bound the main squarefree and balancing terms. Fix a data d -set X . Its residual and restored variable cells lie among the squarefree root tuples. When $d = 2$, condition (A2) permits at most one state index on X , and (11) shows that its state cell is disjoint from the residual and variable cells on every translated support. Thus the total root contribution associated with X is at most $e_k(\mathbf{w})$. Multiplying by $\prod_{z \in X} x_z$ and summing over X bounds all squarefree terms by $E_d(\mathbf{x}) e_k(\mathbf{w})$. Also,

$$e_2(\mathbf{u}^B) = \frac{1}{2} \left(w_B^2 - \sum_a (u_a^B)^2 \right) \leq \frac{q-1}{2q} w_B^2 = \theta_q w_B^2.$$

Thus the balancing templates contribute at most $\theta_q E_d(\mathbf{x}) J_k(\mathbf{w})$.

It remains to bound the positive auxiliary profiles: type, state when $d > 2$, and correction. Fix one translate, let j be its data degree, and let S be its root support, so $|S| = r - j$. Its root cell lies in Ω^S , and hence its root contribution is at most $\prod_{B \in S} w_B$. Its data factor is at most $\rho^j / j!$: for type profiles use $\rho_i^\mu \leq \rho$, for state profiles use $\Delta_\delta^\mu \leq E_2(\mathbf{x}) \leq \rho^2 / 2$, and for the correction profile use $E_{d+1}(\mathbf{x}) \leq \rho^{d+1} / (d+1)!$. Since $\rho + \sum_{B \in S} w_B \leq 1$, AM–GM applied to j copies of ρ/j and the $r-j$ values w_B gives

$$\frac{\rho^j}{j!} \prod_{B \in S} w_B = \frac{j^j}{j!} \left(\frac{\rho}{j} \right)^j \prod_{B \in S} w_B \leq \frac{j^j}{j!} r^{-r}.$$

Equality in this relaxation occurs at $\rho = j/r$, $w_B = 1/r$ for $B \in S$, with all other macroblock weights zero. There are $|I|$ type profiles, h_{st} state profiles, and one correction profile, each with P translates; state terms are included here only when $d > 2$. Their total contribution is therefore at most

$$P \left(|I| \frac{1^1}{1!} + \mathbf{1}_{\{d>2\}} h_{\text{st}} \frac{2^2}{2!} + \frac{(d+1)^{d+1}}{(d+1)!} \right) r^{-r} = \Delta_{r,P}.$$

Finally, $E_d(\mathbf{x}) \leq \rho^d / d!$, proving the second inequality. $\square$

5.3 Dominance of the balancing term and exact root-label uniformity

The global bound now separates low-value weightings from those near the balanced point. Any weighting whose value exceeds $(1 - \eta) L_{\theta_q}$ must lie in the neighborhood U and have enough data mass for the strict fiber-concavity estimate to apply.

Lemma 5.8. *Let (G, μ) be a link-complete rooted graph with a probability weighting ω . With the notation introduced above, at least one of the following alternatives holds.*

1. $p_G(\omega) \leq (1 - \eta)L_{\theta_q}$.
2. One has $\rho_- \leq \rho \leq \rho_+$, $\ell \leq w_B \leq u_+$ for every $B \in \mathbb{Z}_P$, and $E_d(\mathbf{x}) \geq \underline{m}_d$.

Proof. By Lemma 5.7 and (30),

$$p_G(\omega) \leq E_d(\mathbf{x})A_{\theta_q}(\mathbf{w}) + \eta L_-.$$

If the first term is at most $(1 - 2\eta)L_{\theta_q}$, then $L_- \leq L_{\theta_q}$ gives alternative 1. Assume otherwise. Since $E_d(\mathbf{x}) \leq \rho^d/d!$, we have $\Phi_{\theta_q}(\rho, \mathbf{w}) > (1 - 2\eta)L_{\theta_q}$. The global stability bound (19) then gives

$$\|(\rho, \mathbf{w}) - p\|_2^2 < 8 \cdot 6^d r^2 \eta = \frac{1}{8} \varepsilon^2 u_0^2.$$

Hence $|w_B - u_0| < \varepsilon u_0$ for every B and $|\rho - \rho_0| < \varepsilon u_0 \leq \varepsilon \rho_0$. Thus $(\rho, \mathbf{w}) \in U$.

Let $M_{\theta_q} := A_{\theta_q}(P^{-1}, \dots, P^{-1})$. By homogeneity,

$$L_{\theta_q} = \frac{1}{d!} \left(\frac{d}{r}\right)^d \left(\frac{k}{r}\right)^k M_{\theta_q}.$$

It remains to verify the lower bound on $E_d(\mathbf{x})$. Set $s := \sum_B w_B \leq 1$. If $s > 0$, homogeneity and pair averaging give

$$A_{\theta_q}(\mathbf{w}) = s^k A_{\theta_q}(\mathbf{w}/s) \leq s^k M_{\theta_q} \leq M_{\theta_q}.$$

If $s = 0$, the same bound is trivial. Combining it with $E_d(\mathbf{x})A_{\theta_q}(\mathbf{w}) > (1 - 2\eta)L_{\theta_q}$ and $\eta < 1/8$ gives

$$E_d(\mathbf{x}) > \frac{3}{4} \frac{1}{d!} \left(\frac{d}{r}\right)^d \left(\frac{k}{r}\right)^k \geq \underline{m}_d.$$

This establishes alternative 2 and completes the proof. $\square$

In the high-value alternative, strict concavity and translation symmetry now force exact uniformity inside every root macroblock.

Corollary 5.9. *For fixed data weights and macroblock totals, if alternative 2 of Lemma 5.8 holds, then the map $\mathbf{u} \mapsto p_G(\omega_{\mathbf{u}})$ on $\mathcal{X}(\mathbf{w})$ has the unique maximizer*

$$u_a^B = \frac{w_B}{q} \quad \text{for } B \in \mathbb{Z}_P \text{ and } a \in \Omega. \quad (38)$$

Indeed, Corollary 5.6 gives strict concavity on $\mathcal{X}(\mathbf{w})$, while part (iv) of Lemma 4.1 makes every template family invariant under simultaneous diagonal translation of the labels. Averaging over the finite group $(\Omega, +)$ produces (38); concavity and invariance make this orbit average a maximizer, and strict concavity makes it unique.

5.4 Completion of the local declarations

Exact root-label uniformity allows us to complete all local declarations without decreasing the value of the weighted edge polynomial. The repair inequalities become

$$\tau m_T > \sigma \rho_+ M_S + c_v M_W \frac{\rho_+^{d-1}}{(d-1)!} \quad \text{and} \quad (39)$$

$$\sigma m_S > c_v M_W \frac{\rho_+^{d-2}}{(d-2)!}. \quad (40)$$

The first inequality says that adding a missing type bundle pays for deleting all newly incompatible state and variable bundles. The second says the analogous statement for adding a missing state bundle. We use them in that order, and then insert every remaining permitted variable orbit.

Lemma 5.10. *Let (G, μ) be a link-complete admissible rooted graph with a weighting ω satisfying alternative 2 of Lemma 5.8 and (38). There is a link-complete rooted graph (G^{rep}, μ) whose underlying graph belongs to $\mathcal{C}_\dagger$, whose vertex set consists of the same roots and positive-weight data vertices, and whose weighting is the restriction of ω , such that $p_{G^{\text{rep}}}(\omega) \geq p_G(\omega)$ and the following properties hold:*

1. *every positive-weight data vertex carries exactly one type and hence belongs to exactly one sort;*
2. *every data pair has exactly one compatible state or dummy state;*
3. *on every d -set X of composition $\mathbf{d}$, the declared variable parameters are exactly $S_{\mathbf{v}, m_{\beta_X}}$.*

Proof. Let $U^+ := \{z \in U_\mu : x_z > 0\}$. Delete all other data vertices and restrict ω , retaining the same notation. Every removed edge has weight zero. All templates required on the remaining vertices were already present, while deletion can only remove declarations; hence admissibility and link completeness are preserved and $p_G(\omega)$ is unchanged.

Fix a linear order on U^+ and the induced lexicographic orders on its pairs and d -sets. All subsequent graphs use the fixed vertex set $\text{im } \mu \cup U^+$, rooting μ , and weighting ω ; declarations always refer to the current graph.

Every operation below inserts or deletes a complete bundle or orbit. By Lemma 4.3, distinct declarations use disjoint edge sets, all disjoint from the unconditional templates. When $d = 2$, the same conclusion follows from the reserve decomposition (10). Thus deleting a bundle or orbit removes exactly one declaration, and inserting one creates exactly one declaration. Deletion can only relax conditions (A1)–(A5); because all remaining declarations stay complete and the unconditional families are untouched, both admissibility and link completeness are preserved.

The macroblock totals remain fixed. Alternative 2 gives $\rho \leq \rho_+$ and $\ell \leq w_B \leq u_+$, where $\rho := \sum_{z \in U^+} x_z$. When the root-label weights are uniform within each macroblock, a complete type bundle on z contributes $\tau x_z R_{\text{ty}}(\mathbf{w})$, a state bundle on $\{z, v\}$ contributes $\sigma x_z x_v R_{\text{st}}(\mathbf{w})$, and all variable orbits on a fixed d -set X contribute at most $c_v(\prod_{u \in X} x_u) R_{\text{m}}(\mathbf{w})$. The cutoff bounds give $R_{\text{ty}}(\mathbf{w}) \geq m_T$, $m_S \leq R_{\text{st}}(\mathbf{w}) \leq M_S$, and $R_{\text{m}}(\mathbf{w}) \leq M_W$.

We maintain the following invariant throughout: the current rooted graph is admissible and link-complete, its vertices and weighting are fixed, and its value under ω never decreases.

Typing the vertices. Process U^+ in order, leaving already typed vertices unchanged. For an untyped vertex z , choose any sort $i(z) \in I$. Delete every state bundle on a pair $\{z, v\}$ with v already typed whose declared sort pair differs from $\{i(z), i(v)\}$. Next delete every variable orbit on a d -set containing z that would violate condition (A4) or (A5) after assigning type $i(z)$ to z . Finally, insert the complete type- $i(z)$ bundle.

This composite step preserves admissibility. The vertex z receives one type, state uniqueness is unaffected, and every state newly incompatible with the endpoint types was removed. Only d -sets containing z become newly subject to conditions (A4) and (A5), and their forbidden orbits were deleted. The inserted type bundle is complete, so link completeness is preserved as well.

By condition (A2), at most one state bundle is deleted for each $v \neq z$, so their total contribution is at most

$$\sigma x_z R_{\text{st}}(\mathbf{w}) \sum_{v \neq z} x_v \leq \sigma \rho_+ M_S x_z.$$

For each d -set containing z , all its variable orbits contribute no more than the full parameter set $S_{\mathbf{v}, N}$. Hence the deleted variable weight is at most

$$c_v M_W x_z e_{d-1}((x_u)_{u \neq z}) \leq c_v M_W x_z \frac{\rho_+^{d-1}}{(d-1)!}.$$

The new type bundle contributes at least $\tau m_T x_z$, so (39) makes the net change nonnegative.

After all vertex steps, every vertex in U^+ has exactly one type.

Completing the state declarations. Process the data pairs in order. An existing state is already unique and compatible by conditions (A2) and (A3). For an undeclared pair $\{z, v\}$, let $i(z), i(v)$ be its endpoint types. Set $\delta_{zv} := (\{i(z), i(v)\}, 0)$ when this sort pair is visible, and otherwise use the dummy state $\delta_{zv} := (\{i(z), i(v)\}, *)$. Before inserting the complete state- δ_{zv} bundle, delete every variable orbit on a d -set containing $\{z, v\}$ that would violate condition (A5) afterward.

The pair receives exactly one compatible state, no type or composition changes, and every newly forbidden variable orbit has been removed. Thus admissibility and link completeness are preserved.

The inserted state bundle contributes at least $\sigma m_S x_z x_v$, whereas the deleted variable orbits contribute at most

$$c_v M_W x_z x_v e_{d-2}((x_u)_{u \notin \{z, v\}}) \leq c_v M_W x_z x_v \frac{\rho_+^{d-2}}{(d-2)!}.$$

By (40), this step also cannot decrease the value.

After all pair steps, every data pair carries exactly one compatible state or dummy state.

Completing the variable declarations. Process the d -sets of composition $\mathbf{d}$. Each such X is now uniquely typed and has a unique compatible state on every visible pair, so β_X is defined. Condition (A5) gives $\Xi_\mu(X) \subseteq S_{v, m_{\beta_X}}$. Insert the complete variable orbit for every missing parameter in $S_{v, m_{\beta_X}}$. These parameters satisfy both variable conditions by construction; insertion preserves all earlier declarations and link completeness and cannot decrease the value under ω . Consequently, $\Xi_\mu(X) = S_{v, m_{\beta_X}}$ for every d -set X of composition $\mathbf{d}$.

Let G^{rep} be the final graph. The invariant and the initial zero-weight deletion give $p_{G^{\text{rep}}}(\omega) \geq p_G(\omega)$. The rooted graph remains admissible and link-complete, so its underlying graph lies in $\mathcal{C}_f$, and the three asserted completion properties hold. $\square$

5.5 Scalar reduction

After repair, the local declarations define an actual weighted $\mathfrak{M}$ -graph on the data vertices. For $i \in I$, let $V_i := \{z \in U^+ : z \text{ carries type } i\}$. By Lemma 5.10, these sets partition $V_{\text{data}} := U^+$. Put $\rho_i := \sum_{z \in V_i} x_z$ and $\rho := \sum_i \rho_i$.

Let G_{data} be the $\mathfrak{M}$ -graph with vertex sorts $(V_i)_{i \in I}$. On a visible diagonal block $\{i, i\}$, define

$$E_{ii}(G_{\text{data}}) := \left\{ \{z, z'\} \in \binom{V_i}{2} : \{z, z'\} \in \text{St}_\mu^{\{i, i\}, 1} \right\},$$

and on a visible block $\{i, j\}$ with $i < j$, define

$$E_{ij}(G_{\text{data}}) := \left\{ (z, z') \in V_i \times V_j : \{z, z'\} \in \text{St}_\mu^{\{i, j\}, 1} \right\}.$$

Invisible blocks carry no relation; their pairs have the dummy state. The uniqueness and compatibility conclusions of Lemma 5.10 make these relations well defined. Define the injective data factor and its with-replacement analogue by

$$D_{\mathbf{d}}(\mathbf{x}) := \prod_{i \in I} e_{d_i}((x_z)_{z \in V_i}) \quad \text{and} \quad D_{\mathbf{d}}(\boldsymbol{\rho}) := \frac{\prod_i \rho_i^{d_i}}{\nu_{\mathfrak{M}}}.$$

Define the corresponding pattern polynomial by

$$\mathcal{T}_f(\mathbf{x}) := \sum_{\substack{X_i \in \binom{V_i}{d_i} \\ i \in I}} a_{[G_{\text{data}}[\cup_i X_i]]_{\mathfrak{M}}} \prod_{i \in I} \prod_{z \in X_i} x_z.$$

We now identify the variable contribution. Since every data vertex is typed, condition (A4) forces $\Xi_\mu(X) = \emptyset$ unless X has composition $\mathbf{d}$. For a set of the correct composition, Lemma 5.10 gives $\Xi_\mu(X) = S_{v, m_\beta X}$. Because $m_\beta = Na_\beta$, part (ii) of Lemma 4.1 yields

$$\frac{|S_{v, m_\beta}|}{q^3} = \frac{1}{q^2} \frac{|S_{v, m_\beta}|}{q} = \frac{\vartheta_v m_\beta}{q^2 N} = c_v a_\beta.$$

Therefore the variable term in (35) becomes

$$\frac{R_m(\mathbf{w})}{q^3} \sum_{X \in \binom{U^+}{d}} |\Xi_\mu(X)| \prod_{z \in X} x_z = c_v \mathcal{T}_f(\mathbf{x}) R_m(\mathbf{w}).$$

The completed type and state declarations also give

$$\sum_i \rho_i^\mu = \rho \quad \text{and} \quad \sum_\delta \Delta_\delta^\mu = E_2(\mathbf{x}).$$

Substituting these identities into (35) gives

$$\begin{aligned} p_{G^{\text{rep}}}(\omega) &= E_d(\mathbf{x}) C_{\text{res}}(\mathbf{w}) + c_v \mathcal{T}_f(\mathbf{x}) R_m(\mathbf{w}) + \theta_q E_d(\mathbf{x}) J_k(\mathbf{w}) \\ &\quad + \tau \rho R_{\text{ty}}(\mathbf{w}) + \sigma E_2(\mathbf{x}) R_{\text{st}}(\mathbf{w}) + \gamma E_{d+1}(\mathbf{x}) R_{\text{cor}}(\mathbf{w}). \end{aligned} \quad (41)$$

To remove the remaining dependence on the individual data vertices, let $\Delta := \{(\rho, \mathbf{w}) \geq 0 : \sum_i \rho_i + \sum_B w_B = 1\}$. For $t \in [0, 1]$, define H_t on Δ by

$$\begin{aligned} H_t(\rho, \mathbf{w}) &:= \frac{\rho^d}{d!} C_{\text{res}}(\mathbf{w}) + c_v t D_{\mathbf{d}}(\rho) R_m(\mathbf{w}) + \theta_q \frac{\rho^d}{d!} J_k(\mathbf{w}) \\ &\quad + \tau \rho R_{\text{ty}}(\mathbf{w}) + \sigma \frac{\rho^2}{2} R_{\text{st}}(\mathbf{w}) + \gamma \frac{\rho^{d+1}}{(d+1)!} R_{\text{cor}}(\mathbf{w}). \end{aligned} \quad (42)$$

The polynomial H_t is obtained from (41) by replacing injective data factors with their with-replacement bounds and the normalized matrix evaluation by t . On the sort-balanced slice $\rho_i = (d_i/d)\rho$, it coincides with $\hat{H}_t$. In particular, at $\rho_i = d_i/r$ and $w_B = k/(rP)$, its value is $\hat{H}_t(p)$.

Proposition 5.11. *For $t := \text{Val}_{\mathfrak{M}}(f)$,*

$$\Lambda(\mathcal{C}_f) = \max_{(\rho, \mathbf{w}) \in \Delta} H_t(\rho, \mathbf{w}). \quad (43)$$

Proof. Every sampling evaluation is the expectation of a coefficient in $[0, 1]$, so $0 \leq t \leq 1$. We prove the two inequalities in (43) separately.

For the upper bound, fix $G_0 \in \mathcal{C}_f$ and a probability weighting ω . By Lemma 5.1, embed G_0 in a link-complete $G \in \mathcal{C}_f$ and give all new vertices weight zero. Then $p_{G_0}(\omega) \leq p_G(\omega)$.

If alternative 1 of Lemma 5.8 holds, then $p_G(\omega) \leq (1 - \eta) L_{\theta_q}$, whereas (O1) gives

$$\max_{(\rho, \mathbf{w}) \in \Delta} H_t(\rho, \mathbf{w}) \geq \hat{H}_t(p) \geq (1 - \eta) L_{\theta_q}.$$

If alternative 2 holds, root-label uniformization and semantic repair do not decrease the value, so we may use (41). Since C_{res} is nonnegative, $E_d(\mathbf{x}) C_{\text{res}}(\mathbf{w}) \leq \rho^d C_{\text{res}}(\mathbf{w})/d!$; the analogous bounds hold for E_2 and E_{d+1} . Also $D_{\mathbf{d}}(\mathbf{x}) \leq D_{\mathbf{d}}(\rho)$. If some $\rho_i = 0$, both $\mathcal{T}_f(\mathbf{x})$ and $D_{\mathbf{d}}(\rho)$ vanish because every d_i is positive. Otherwise, normalize the weights within each sort and apply Lemma 2.1 to get $\mathcal{T}_f(\mathbf{x}) \leq t D_{\mathbf{d}}(\rho)$. Substitution proves the upper bound in the second alternative as well.

For the reverse inequality, fix $(\rho, \mathbf{w}) \in \Delta$. Let (A_n) be the uniformly weighted finite $\mathfrak{M}$ -graphs from Lemma 2.1. Give every vertex of sort i weight $\rho_i/|V_i(A_n)|$ and every root label in W_B weight w_B/q .

Build a link-complete rooted graph by inserting all unconditional templates, the type bundle determined by each data sort, and the state bundle recording each visible relation entry; use dummy-state bundles on invisible blocks. On every d -set of composition $\mathbf{d}$, insert exactly the variable orbits permitted by its induced pattern. The resulting rooted graph is admissible and link-complete. Denote its data-weight vector by $\mathbf{x}^{(n)}$.

Since $|V_i(A_n)| \rightarrow \infty$ in every sort, the largest data-vertex weight tends to zero. For any nonnegative vector $\mathbf{y}$ of total weight s and largest coordinate δ , a union bound on repeated indices gives, for $j \geq 1$,

$$0 \leq s^j - j!e_j(\mathbf{y}) \leq \binom{j}{2} \delta s^{j-1}.$$

Apply this estimate to the full data vector and separately within each sort. Then $E_j(\mathbf{x}^{(n)}) \rightarrow \rho^j/j!$ for $j \in \{2, d, d+1\}$ and $D_{\mathbf{d}}(\mathbf{x}^{(n)}) \rightarrow D_{\mathbf{d}}(\rho)$. Moreover, $\mathcal{T}_{\mathfrak{f}}(\mathbf{x}^{(n)}) = (\prod_i \rho_i^{d_i}) T_{\mathfrak{f}}^{\mathfrak{M}}(A_n, \text{unif}) \rightarrow tD_{\mathbf{d}}(\rho)$. Equation (41) therefore converges to $H_t(\rho, \mathbf{w})$. Taking the supremum over these constructions and then the maximum over Δ proves the reverse inequality. The maximum exists because H_t is continuous on the compact simplex Δ . $\square$

5.6 A common maximizer for the reduced polynomials

It remains to maximize H_t . There are two levels of balance. Weighted AM–GM first fixes the split of the data mass among the sorts. On that slice, the global stability of Φ_{θ_q} dominates the small perturbation Q_t , whose gradient vanishes at p by the choice of the correction parameter. Thus the same point maximizes H_t for every $t \in [0, 1]$.

Lemma 5.12. *Fix $t \in [0, 1]$ and $(\rho, \mathbf{w}) \in \Delta$, and set $\rho := \sum_{i \in I} \rho_i$. Consider the balanced sort masses*

$$\rho_i = \frac{d_i}{d} \rho \quad \text{for every } i \in I. \quad (44)$$

Replacing the original sort masses by these values does not decrease H_t . If $tR_m(\mathbf{w}) > 0$, then the balanced vector is the unique choice of sort masses that maximizes H_t . At the balanced vector, $D_{\mathbf{d}}(\rho) = \zeta_{\mathfrak{M}} \rho^d/d!$.

This follows directly from weighted AM–GM. Only the variable term depends on the individual sort masses, and its coefficient is nonnegative; hence $\prod_i \rho_i^{d_i}$ is maximized at (44). The maximizer is unique when the coefficient is positive, while H_t is independent of the sort split when it vanishes. Direct substitution gives the stated value of $D_{\mathbf{d}}$.

We next verify stationarity at the balanced point. Consider a general profile of data degree j : for $0 \leq j \leq r$, let $a_j \in \mathbb{R}$, let $R_{r-j}(\mathbf{w})$ be cyclically invariant and homogeneous of degree $r-j$, and set

$$P_j(\rho, \mathbf{w}) := \frac{a_j}{j!} \rho^j R_{r-j}(\mathbf{w}).$$

At $\mathbf{w}_0 = u_0 \mathbf{1}$, cyclic invariance makes all partial derivatives of R_{r-j} equal. Euler's identity therefore gives, for every B ,

$$Pu_0 \partial_{w_B} R_{r-j}(\mathbf{w}_0) = \sum_{C \in \mathbb{Z}_P} w_{0,C} \partial_{w_C} R_{r-j}(\mathbf{w}_0) = (r-j) R_{r-j}(\mathbf{w}_0).$$

Using $\rho_0 = d/r$ and $Pu_0 = k/r$, we obtain

$$\partial_{\rho} P_j(p) = \frac{j}{\rho_0} P_j(p) = \frac{jr}{d} P_j(p), \quad \partial_{w_B} P_j(p) = \frac{r-j}{Pu_0} P_j(p) = \frac{r(r-j)}{k} P_j(p).$$

Hence

$$\partial_\rho P_j(p) - \partial_{w_B} P_j(p) = \left(\frac{jr}{d} - \frac{r(r-j)}{k} \right) P_j(p) = \frac{r^2(j-d)}{dk} P_j(p) \quad \text{for every } B \in \mathbb{Z}_P. \quad (45)$$

Thus every profile of data degree d is stationary on $T\Delta_{\text{mac}}$ at p .

The only remaining profiles have data degrees 1, 2, and $d+1$. Apply (45) to

$$P_1 := \tau \rho R_{\text{ty}}, \quad P_2 := \sigma \frac{\rho^2}{2} R_{\text{st}}, \quad P_{d+1} := \gamma \frac{\rho^{d+1}}{(d+1)!} R_{\text{cor}}.$$

For every $B \in \mathbb{Z}_P$,

$$\begin{aligned} \frac{dk}{r^2} (\partial_\rho - \partial_{w_B})(P_1 + P_2 + P_{d+1})(p) = & - (d-1) \tau \rho_0 R_{\text{ty}}(\mathbf{w}_0) \\ & - (d-2) \sigma \frac{\rho_0^2}{2} R_{\text{st}}(\mathbf{w}_0) + \gamma \frac{\rho_0^{d+1}}{(d+1)!} R_{\text{cor}}(\mathbf{w}_0) = 0, \end{aligned}$$

where the last equality is exactly the correction identity (25). Every other summand of Q_t has data degree d , so altogether

$$DQ_t(p)[v] = 0 \quad \text{for } t \in [0, 1] \text{ and } v \in T\Delta_{\text{mac}}. \quad (46)$$

The global stability estimate, the perturbation bound, and this stationarity identity together identify the common maximizer.

Proposition 5.13. *The point p uniquely maximizes $\hat{H}_t$ on Δ_{mac} for every $t \in [0, 1]$. Let $p_{\text{full}} \in \Delta$ be the point determined by*

$$\rho_i = \frac{d_i}{r} \quad \text{for every } i \in I \quad \text{and} \quad w_B = \frac{k}{rP} \quad \text{for every } B \in \mathbb{Z}_P.$$

The point p_{full} maximizes H_t on Δ . This maximizer is unique when $t > 0$; for $t = 0$, only the split of the total data mass among the sorts may be nonunique.

Proof. On the sort-balanced slice, H_t becomes $\hat{H}_t = \Phi_{\theta_q} + Q_t$. Fix $x \in \Delta_{\text{mac}} \setminus \{p\}$ and put $h := x - p$. The global anchor stability gives

$$\Phi_{\theta_q}(p) - \Phi_{\theta_q}(x) \geq \mu_{\text{mac}} \|h\|_2^2.$$

Taylor's theorem along $[p, x]$, together with the stationarity (46) and Hessian bound (O2), gives

$$|Q_t(x) - Q_t(p)| < \frac{1}{2} \mu_{\text{mac}} \|h\|_2^2.$$

The anchor loss therefore dominates the perturbation:

$$\hat{H}_t(p) - \hat{H}_t(x) > \frac{1}{2} \mu_{\text{mac}} \|h\|_2^2.$$

The right-hand side is positive, so p uniquely maximizes $\hat{H}_t$. Applying Lemma 5.12 lifts this conclusion to p_{full} on Δ . Since $R_m(\mathbf{w}_0) > 0$, the sort split is unique when $t > 0$. When $t = 0$, H_0 is independent of that split, so nonuniqueness occurs precisely when $|I| > 1$. $\square$

Proof of Lemma 3.2. Set $h_{\mathfrak{M}, r, N} := H_0(p_{\text{full}})$ and

$$b_{\mathfrak{M}, r, N} := c_v \frac{\prod_i (d_i/r)^{d_i}}{\nu_{\mathfrak{M}}} R_m \left(\frac{k}{rP}, \dots, \frac{k}{rP} \right). \quad (47)$$

The polynomial H_0 has effectively computable rational coefficients and p_{full} is rational, so $h_{\mathfrak{M},r,N} \in \mathbb{Q}$ is effectively computable. Formula (47) likewise shows that $b_{\mathfrak{M},r,N} \in \mathbb{Q}_{>0}$ is effectively computable. Both depend only on $\mathfrak{M}$, r , and N .

Propositions 5.11 and 5.13 give

$$\Lambda(\mathcal{C}_f) = H_0(p_{\text{full}}) + b_{\mathfrak{M},r,N} \text{Val}_{\mathfrak{M}}(f).$$

This is (3), completing the proof. $\square$

6 Consequences of the compiler theorem

We first give a short graphon application, reducing the Sidorenko property to equality of two Turán densities. We then prove the two main results: undecidability at rational thresholds and the existence of transcendental Turán densities. In each case, the main task is to construct a fixed-order sampling input with the required value; the compiler theorem then transfers that value through a strictly increasing rational affine map.

6.1 A graphon warm-up: Sidorenko equality

Proof of Corollary 1.5. Let $e := e(H)$ and put $D := D(n)$, where

$$D(n) := \max \left\{ 2, n + 2 \left\lfloor \frac{n^2}{4} \right\rfloor \right\}.$$

Since H is bipartite, $e \leq \lfloor n^2/4 \rfloor$. Among the D labeled sampled positions, reserve one position for each vertex of H and two further positions for each edge, denoted by

$$x_1, y_1, \dots, x_e, y_e;$$

any remaining positions are unused. For a graph G on these positions, let $G_{uv} \in \{0, 1\}$ denote the adjacency indicator and set

$$X_H(G) := \prod_{uv \in E(H)} G_{uv}, \quad X_e(G) := \prod_{j=1}^e G_{x_j y_j},$$

where an empty product is 1. Define

$$b_H(G) := \frac{1}{2} + \frac{1}{2}(X_e(G) - X_H(G)).$$

Thus $b_H(G) \in \{0, 1/2, 1\}$.

To obtain coefficients indexed by unlabeled graphs, symmetrize this statistic. For $\beta \in \mathcal{U}_D$, choose a representative G on the labeled positions and set

$$a_{\beta}^H := \frac{1}{D!} \sum_{\sigma \in S_D} b_H(G^{\sigma}), \quad (G^{\sigma})_{uv} := G_{\sigma(u)\sigma(v)}. \quad (48)$$

Relabeling G only permutes the summands, so a_{β}^H is well defined, effectively computable, and belongs to

$$[0, 1] \cap \frac{1}{2D!} \mathbb{Z}.$$

Let $a_{\beta}^0 := 1/2$ for every $\beta \in \mathcal{U}_D$, and use the common denominator $N := 2D!$.

For every graphon W , the sampled graph $\mathbb{G}(D, W)$ has a permutation-invariant distribution. Hence

$$\mathbb{E} a_{[\mathbb{G}(D, W)]}^H = \mathbb{E} b_H(\mathbb{G}(D, W)).$$

The pairs $x_j y_j$ have disjoint endpoints, so independence of their latent variables and edge trials gives

$$\mathbb{E} X_e(\mathbb{G}(D, W)) = p(W)^e.$$

Similarly, the definition of homomorphism density gives

$$\mathbb{E} X_H(\mathbb{G}(D, W)) = t(H, W).$$

The unused sampled variables integrate out in both identities. Therefore

$$\mathbb{E} a_{[\mathbb{G}(D, W)]}^H = \frac{1}{2} + \frac{1}{2}(p(W)^e - t(H, W)), \quad (49)$$

whereas $\mathbb{E} a_{[\mathbb{G}(D, W)]}^0 = 1/2$. Taking suprema over graphons yields

$$\text{Val}_D^{\text{gr}}(a^H) = \frac{1}{2} + \frac{1}{2}\Delta_H^{\text{gr}}, \quad \text{Val}_D^{\text{gr}}(a^0) = \frac{1}{2}. \quad (50)$$

If $e = 0$, then $p(W)^e = t(H, W) = 1$ for every W . If $e \geq 1$, the zero graphon makes their difference zero. Thus $\Delta_H^{\text{gr}} \geq 0$, and

$$\Delta_H^{\text{gr}} = 0 \iff H \text{ has the Sidorenko property.} \quad (51)$$

Set

$$r_{\text{Sid}}(n) := \max\{3, r_{\text{aff}}(D(n))\}.$$

Apply [Corollary 1.4](#) to a^H and a^0 with the same sampling order D , target uniformity r , and common denominator $N = 2D!$. Since the affine constants depend only on D , r , and N , the two applications have the same constants $A \in \mathbb{Q}$ and $B \in \mathbb{Q}_{>0}$. The resulting families satisfy

$$\pi(\mathcal{F}_{H,r}) = A + B \left(\frac{1}{2} + \frac{1}{2}\Delta_H^{\text{gr}} \right), \quad \pi(\mathcal{F}_{H,r}^0) = A + \frac{B}{2}.$$

Their difference is $B\Delta_H^{\text{gr}}/2$. Since $B > 0$, (51) gives the claimed equivalence. Finally, r_{Sid} is effectively computable because r_{aff} is. $\square$

6.2 Undecidability at rational thresholds

We prove [Theorem 1.1](#) in three stages. First, a fixed Diophantine representation encodes membership in a computably enumerable undecidable set. Next, the Hatami–Norine edge–triangle penalty converts integer solutions into negativity of a polynomial on a product of edge–triangle regions. Finally, one graph sort for each Diophantine variable realizes that polynomial as a fixed-order sampling input. After normalization, the resulting matrix value is $1/2$ off the undecidable set and strictly larger than $1/2$ on it.

6.2.1 A fixed Diophantine representation

Fix a computably enumerable undecidable set $\mathcal{S} \subseteq \mathbb{N}$ together with an enumerator. Finite-variable Diophantine reductions of Matiyasevich and Robinson [32], the Davis–Putnam–Robinson–Matiyasevich theorem [8, 31], and the universal form of Jones [22, p. 549] yield the uniform representation needed below.

Recall that an *acceptable numbering* $(\varphi_e)_e$ is a universal effective enumeration of the partial computable functions with the parameterization property. Its domains $W_e := \text{dom } \varphi_e$ form an

acceptable numbering of the computably enumerable sets. Thus, from an effective description of a computably enumerable relation $R(\mathbf{u}, a)$ and a parameter $\mathbf{u}$, one can compute an index $s(\mathbf{u})$ satisfying $W_{s(\mathbf{u})} = \{a : R(\mathbf{u}, a)\}$.

We use the following fixed-variable universal Diophantine representation.

Theorem 6.1. *One can fix an integer $m \geq 1$, a polynomial $U \in \mathbb{Z}[V, A, Z_1, \dots, Z_m]$, and an acceptable numbering $(W_v)_{v \in \mathbb{N}}$ of the computably enumerable subsets of $\mathbb{N}$ such that, for all $v, a \in \mathbb{N}$,*

$$a \in W_v \iff \exists(n_1, \dots, n_m) \in \mathbb{N}^m \text{ such that } U(v, a, n_1, \dots, n_m) = 0.$$

Proof. Let $\mathbb{N}_0 := \{0, 1, 2, \dots\}$. Fix an acceptable numbering $(\varphi_e)_{e \in \mathbb{N}_0}$ and write $\widetilde{W}_e := \text{dom } \varphi_e$. Its universal membership relation

$$K := \{(e, a) \in \mathbb{N}_0^2 : a \in \widetilde{W}_e\}$$

is computably enumerable. We use the effective form of the constructive DPRM theorem: from an enumerator for a computably enumerable relation $R \subseteq \mathbb{N}_0^s$, one can compute an integer $\ell \geq 1$ and the finite coefficient list of a polynomial

$$P_R(X_1, \dots, X_s, Y_1, \dots, Y_\ell) \in \mathbb{Z}[\mathbf{X}, \mathbf{Y}]$$

such that

$$\mathbf{u} \in R \iff \exists \mathbf{y} \in \mathbb{N}_0^\ell \text{ such that } P_R(\mathbf{u}, \mathbf{y}) = 0.$$

This follows by tracing the constructive reductions in Matiyasevich [33, Sections 4.5 and 5.5]; see also the fixed universal form of Jones [22, p. 549]. Apply the procedure to a fixed enumerator of K . We obtain an integer $m \geq 1$ and an effectively computable polynomial $F(E, A, Y_1, \dots, Y_m) \in \mathbb{Z}[E, A, Y_1, \dots, Y_m]$ such that

$$a \in \widetilde{W}_e \iff \exists(y_1, \dots, y_m) \in \mathbb{N}_0^m \text{ such that } F(e, a, y_1, \dots, y_m) = 0.$$

Both m and the coefficient list of F are fixed. Define

$$W_v := \{a \in \mathbb{N} : a - 1 \in \widetilde{W}_{v-1}\} \quad \text{and} \quad U(V, A, Z_1, \dots, Z_m) := F(V - 1, A - 1, Z_1 - 1, \dots, Z_m - 1).$$

The shifted numbering remains acceptable, and the coefficient list of U is computable. The stated equivalence follows from the bijection $n \mapsto n - 1$ between $\mathbb{N}$ and $\mathbb{N}_0$. $\square$

We now specialize the universal representation to $\mathcal{S}$. Since $(W_v)_{v \in \mathbb{N}}$ is acceptable and $\mathcal{S}$ is computably enumerable, its fixed enumerator effectively yields an index $v_{\mathcal{S}}$ such that $W_{v_{\mathcal{S}}} = \mathcal{S}$. Apply Theorem 6.1 with this index and set

$$P(A, Z_1, \dots, Z_m) := U(v_{\mathcal{S}}, A, Z_1, \dots, Z_m).$$

The polynomial P is fixed and effectively computable, and for every $a \in \mathbb{N}$,

$$a \in \mathcal{S} \iff \exists \mathbf{n} \in \mathbb{N}^m \text{ such that } P(a, \mathbf{n}) = 0.$$

For each $a \in \mathbb{N}$, set $Q_a(\mathbf{Z}) := 2P(a, \mathbf{Z})^2 - 1$. Since $P(a, \mathbf{n})$ is integral,

$$Q_a(\mathbf{n}) < 0 \iff P(a, \mathbf{n}) = 0. \tag{52}$$

The number of variables and the degree bound are independent of a ; only the integer coefficients vary.

6.2.2 The edge–triangle penalty

For a finite graph F and a probability-weighted finite graph (H, ω) , let $t(F; H, \omega)$ be the probability that an independently ω -sampled map $V(F) \rightarrow V(H)$ is a graph homomorphism. We write $t(F; H)$ when the weighting is understood.

We use the edge–triangle penalty of Hatami and Norine [21, Lemma 5.4]. Put $g_{\text{et}}(x) := 2x^2 - x$. For $t \in \mathbb{N}$, define

$$\ell_t(x) := \frac{3t^2 - t - 2}{t(t+1)}x - \frac{2(t-1)}{t+1},$$

and let $L(1) := 1$ and $L(x) := \ell_t(x)$ for $1 - 1/t \leq x < 1 - 1/(t+1)$. These intervals partition $[0, 1]$, adjacent formulas agree at their common endpoint, and

$$\ell_t(1 - 1/t) = g_{\text{et}}(1 - 1/t) \longrightarrow 1.$$

Hence L is continuous on $[0, 1]$, and the *edge–triangle region*

$$\mathcal{R}_{\text{et}} := \{(x, y) \in [0, 1]^2 : y \geq L(x)\}$$

is closed. In the normalization of Hatami and Norine [21, Theorem 5.2], the edge–triangle theorem of Bollobás [4] places the edge and triangle homomorphism densities of every finite graph in $\mathcal{R}_{\text{et}}$. Rational vertex weights are realized by blowups, and approximation together with closedness extends the conclusion to arbitrary real weights. Thus every finite weighted graph H satisfies

$$(t(K_2; H), t(K_3; H)) \in \mathcal{R}_{\text{et}}. \quad (53)$$

Since g_{et} is convex and each ℓ_t is the chord joining consecutive points of its graph, $L(x) \geq g_{\text{et}}(x)$, with equality at every $x = 1 - 1/n$.

The next lemma uses this equality set to convert the fixed Diophantine representation into a uniformly bounded-degree polynomial sign problem on $\mathcal{R}_{\text{et}}^k$.

Lemma 6.2. *There exist a fixed integer $k \geq 1$, a fixed integer $\delta_* \geq 2$, and a computable sequence of integer polynomials $q_a(x_1, \dots, x_k, y_1, \dots, y_k)$ with $\deg q_a \leq \delta_*$ such that*

$$a \notin \mathcal{S} \iff q_a(\mathbf{x}, \mathbf{y}) \geq 0 \text{ whenever } (x_i, y_i) \in \mathcal{R}_{\text{et}} \text{ for every } i \in [k] \text{ and} \quad (54)$$

$$a \in \mathcal{S} \iff q_a\left(\left(1 - \frac{1}{n_i}\right)_{i=1}^k, \left(g_{\text{et}}\left(1 - \frac{1}{n_i}\right)\right)_{i=1}^k\right) < 0 \text{ for some } (n_1, \dots, n_k) \in \mathbb{N}^k. \quad (55)$$

Proof. Take $k := m$ and fix an upper bound $D \geq 1$ on $\deg Q_a$. We first transfer the integer grid to the distinguished points of the edge–triangle boundary. Put

$$p_a(\mathbf{x}) := \prod_{i=1}^k (1 - x_i)^D Q_a\left(\frac{1}{1 - x_1}, \dots, \frac{1}{1 - x_k}\right).$$

This is an integer polynomial of degree at most kD . Moreover, $Q_a \not\equiv 0$, and clearing denominators after the invertible substitutions $Z_i = (1 - x_i)^{-1}$ cannot annihilate it. Hence $p_a \not\equiv 0$, and (52) becomes

$$a \in \mathcal{S} \iff p_a(\mathbf{x}) < 0 \text{ for some } \mathbf{x} \in \{1 - 1/n : n \in \mathbb{N}\}^k. \quad (56)$$

We now add the edge–triangle penalty. For a polynomial p , let $\|p\|_1$ be the sum of the absolute values of its coefficients, and put

$$M_a^0 := 100 \deg(p_a) \|p_a\|_1, \quad M_a := 100 \max\{1, \deg p_a\} \|p_a\|_1.$$

For $M \geq 0$, define

$$q_{a,M}(\mathbf{x}, \mathbf{y}) := p_a(\mathbf{x}) \prod_{i=1}^k (1 - x_i)^6 + M \sum_{i=1}^k (y_i - g_{\text{et}}(x_i)),$$

and set $q_a := q_{a,M_a}$. With $\delta_* := \max\{kD + 6k, 2\}$, we have $\deg q_a \leq \delta_*$ for every a .

We first dispose of the constant case. If $p_a \equiv c$, then $y_i \geq L(x_i) \geq g_{\text{et}}(x_i)$ throughout $\mathcal{R}_{\text{et}}^k$. Thus q_a is nonnegative when $c \geq 0$. If $c < 0$, choose arbitrary $n_1, \dots, n_k \in \mathbb{N}$ and put

$$x_i := 1 - \frac{1}{n_i} \quad \text{and} \quad y_i := g_{\text{et}}(x_i) = L(x_i).$$

At this point the penalty vanishes and $\prod_i (1 - x_i)^6 = \prod_i n_i^{-6} > 0$, so $q_a < 0$. Together with (56), this proves both assertions when $\deg p_a = 0$.

Now suppose that $\deg p_a \geq 1$, so $M_a = M_a^0$. The Hatami–Norine penalty lemma [21, Lemma 5.4], applied to p_a , gives the following equivalence when every $x_i < 1$. Allowing $x_i = 1$ creates no additional negative points: the first term then vanishes, while every penalty term remains nonnegative. Therefore

$$q_{a,M_a^0}(\mathbf{x}, \mathbf{y}) < 0 \text{ for some } (\mathbf{x}, \mathbf{y}) \in \mathcal{R}_{\text{et}}^k \iff p_a(\mathbf{u}) < 0 \text{ for some } \mathbf{u} \in \{1 - 1/n : n \in \mathbb{N}\}^k.$$

Since $q_a = q_{a,M_a^0}$ in this case, (56) completes the proof. $\square$

6.2.3 A fixed multisorted input and the reduction

The preceding lemma detects Diophantine feasibility by the sign of a polynomial on k independent edge–triangle regions. We next realize this polynomial as a fixed-order induced-density input: one graph sort records each edge–triangle pair, while one auxiliary sort supplies a nonnegative factor that can also vanish.

Let $(q_a)_a$, k , and δ_* be as in Lemma 6.2. Put $D_* := 3\delta_*$ and $I_* := \{0, 1, \dots, k\}$. Define the fixed matrix sampling scheme $\mathfrak{D} := (I_*, \mathbf{d}_*, \mathcal{R}_*)$ by

$$d_0 := 4, \quad d_i := D_* \quad (1 \leq i \leq k), \quad \mathcal{R}_* := \{\{i, i\} : i \in I_*\}.$$

Thus each sort induces an ordinary weighted graph, and there are no relations between different sorts. Sort 0 is auxiliary; sorts $1, \dots, k$ encode the edge–triangle variables. The total sampled order

$$d_* := d(\mathfrak{D}) = 4 + kD_*$$

is an absolute constant.

For a finite weighted $\mathfrak{D}$ -graph (G, ω) , let G_i denote its restriction to sort i , and set

$$x_i := t(K_2; G_i), \quad y_i := t(K_3; G_i) \quad (1 \leq i \leq k), \quad z := t(K_2; G_0).$$

We first realize the polynomial $q_a(\mathbf{x}, \mathbf{y})z(1 - z)$ as a quantum graph on this fixed sampling scheme.

Lemma 6.3. *For each a , one can effectively construct a $\mathfrak{D}$ -quantum graph of the form $h_a = \sum_{\beta \in \mathcal{U}_{\mathfrak{D}}} b_{a,\beta} \mathbf{I}_{\mathfrak{D}}(\beta)$ with $b_{a,\beta} \in \nu_{\mathfrak{D}}^{-1} \mathbb{Z}$ for every $\beta \in \mathcal{U}_{\mathfrak{D}}$. On every finite weighted $\mathfrak{D}$ -graph (G, ω) , it satisfies*

$$\widehat{t}_{\mathfrak{D}}(h_a; G, \omega) = q_a(x_1, \dots, x_k, y_1, \dots, y_k) z(1 - z). \quad (57)$$

Proof. Fix a monomial $\prod_i x_i^{\alpha_i} y_i^{\beta_i}$ of q_a . In sort i , take the disjoint union of α_i copies of K_2 , β_i copies of K_3 , and enough isolated vertices to reach D_* vertices. This is possible because

$$2\alpha_i + 3\beta_i \leq 3(\alpha_i + \beta_i) \leq 3\delta_* = D_*.$$

Its weighted homomorphism density is $x_i^{\alpha_i} y_i^{\beta_i}$. In the auxiliary sort, the difference

$$(K_2 \dot{\cup} 2K_1) - (K_2 \dot{\cup} K_2)$$

represents $z - z^2 = z(1 - z)$, with both terms using four vertices. Taking the product over the sorts and then the integer linear combination prescribed by q_a gives a finite quantum graph on the required sampled vertex sets whose evaluation is the right-hand side of (57).

It remains to expand this homomorphism-density expression in the induced basis. Applying (2) to every sampled $\mathfrak{D}$ -graph in the preceding integer linear combination gives coefficients in $\nu_{\mathfrak{D}}^{-1}\mathbb{Z}$ without changing its evaluation. Every step is finite and effective. $\square$

The sign characterization from Lemma 6.2 now translates directly into the infimum of this quantum-graph evaluation.

Lemma 6.4. *For every a , the quantum graph h_a from Lemma 6.3 satisfies*

$$a \notin \mathcal{S} \iff \inf_{G, \omega} \widehat{t}_{\mathfrak{D}}(h_a; G, \omega) = 0 \quad \text{and} \quad (58)$$

$$a \in \mathcal{S} \iff \inf_{G, \omega} \widehat{t}_{\mathfrak{D}}(h_a; G, \omega) < 0. \quad (59)$$

Here both infima range over all finite weighted $\mathfrak{D}$ -graphs (G, ω) .

Proof. By (53), each (x_i, y_i) lies in $\mathcal{R}_{\text{et}}$. If $a \notin \mathcal{S}$, then (54) and $0 \leq z \leq 1$ make (57) nonnegative. Taking an edgeless one-vertex graph in the auxiliary sort gives $z = 0$, so the infimum is exactly zero.

If $a \in \mathcal{S}$, choose $n_1, \dots, n_k$ from (55) and take the uniformly weighted graph K_{n_i} in sort i . Then

$$t(K_2; K_{n_i}) = 1 - \frac{1}{n_i}, \quad t(K_3; K_{n_i}) = \frac{(n_i - 1)(n_i - 2)}{n_i^2} = g_{\text{et}}(1 - 1/n_i).$$

In the auxiliary sort, a uniformly weighted K_2 gives $z = 1/2$. Equation (57) is therefore strictly negative. $\square$

We normalize the coefficients before applying the compiler. Put $\mathbf{1}_{\mathfrak{D}} := \sum_{\beta \in \mathcal{U}_{\mathfrak{D}}} \mathbf{I}_{\mathfrak{D}}(\beta)$, whose evaluation is identically 1. For each a , let

$$C_a := 1 + \max_{\beta \in \mathcal{U}_{\mathfrak{D}}} [|b_{a, \beta}|]$$

and define

$$g_a := \frac{1}{2} \mathbf{1}_{\mathfrak{D}} - \frac{1}{2C_a} h_a. \quad (60)$$

Every induced-density coefficient of g_a lies strictly between 0 and 1, and

$$\text{Val}_{\mathfrak{D}}(g_a) = \frac{1}{2} - \frac{1}{2C_a} \inf_{G, \omega} \widehat{t}_{\mathfrak{D}}(h_a; G, \omega). \quad (61)$$

The integer $N_a := 2C_a \nu_{\mathfrak{D}}$ is an even common denominator computable from a .

The scheme and its sampled order are fixed; only the coefficient list and common denominator vary effectively with a . The preceding two lemmas immediately yield the following summary.

Theorem 6.5. *There is a fixed finite binary matrix sampling scheme $\mathfrak{D}$ of total sampled order d_* , together with a computable sequence of rational inputs g_a having computable even common denominators N_a . Every induced-density coefficient of g_a lies strictly between 0 and 1, and*

$$\text{Val}_{\mathfrak{D}}(g_a) \geq \frac{1}{2}, \quad \text{Val}_{\mathfrak{D}}(g_a) = \frac{1}{2} \iff a \notin \mathcal{S}.$$

Proof of Theorem 1.1. Let $\mathfrak{D}$, d_* , g_a , and N_a be as in Theorem 6.5, and put

$$r_0^{\text{rat}} := \max\{3, r_{\text{aff}}(d_*)\}.$$

Fix $r \geq r_0^{\text{rat}}$ and apply Theorem 1.3 to $(\mathfrak{D}, r, N_a, g_a)$. Let $\mathcal{F}_a$ be the resulting family, and abbreviate

$$A_a := A_{\mathfrak{D}, r, N_a}, \quad B_a := B_{\mathfrak{D}, r, N_a}, \quad \alpha_a := A_a + \frac{B_a}{2}.$$

These numbers are computable and rational, with $B_a > 0$. If $t_a := \text{Val}_{\mathfrak{D}}(g_a)$, then the compiler identity gives $\pi(\mathcal{F}_a) = A_a + B_a t_a \geq \alpha_a$. By Theorem 6.5,

$$\begin{aligned} a \notin \mathcal{S} &\iff t_a = \frac{1}{2} \iff \pi(\mathcal{F}_a) = \alpha_a \iff \pi(\mathcal{F}_a) \leq \alpha_a \quad \text{and} \\ a \in \mathcal{S} &\iff t_a > \frac{1}{2} \iff \pi(\mathcal{F}_a) > \alpha_a. \end{aligned}$$

An oracle for $\leq$ would detect the complement of $\mathcal{S}$ and hence decide $\mathcal{S}$, proving (i). Moreover, every constructed instance satisfies the stronger promise $\pi(\mathcal{F}_a) \geq \alpha_a$, and the same equivalences establish the equality and strict-inequality variants stated after the theorem.

For (ii), use the constant input

$$g^0 := \frac{1}{2} \mathbf{1}_{\mathfrak{D}}.$$

Its sampling evaluation, and hence its matrix value, is identically $1/2$. Since N_a is even, g^0 is a valid input with the same denominator. Apply the compiler to $(\mathfrak{D}, r, N_a, g^0)$ and call the resulting family $\mathcal{G}_a$. The scheme, uniformity, and denominator agree with those used for $\mathcal{F}_a$, so the affine constants are again A_a and B_a . Consequently,

$$\pi(\mathcal{G}_a) = A_a + \frac{B_a}{2} = \alpha_a,$$

and therefore

$$\begin{aligned} \pi(\mathcal{F}_a) &\geq \pi(\mathcal{G}_a), \\ a \notin \mathcal{S} &\iff \pi(\mathcal{F}_a) = \pi(\mathcal{G}_a) \iff \pi(\mathcal{F}_a) \leq \pi(\mathcal{G}_a), \\ a \in \mathcal{S} &\iff \pi(\mathcal{F}_a) > \pi(\mathcal{G}_a). \end{aligned}$$

Hence an oracle for $\leq$ would again decide $\mathcal{S}$, proving (ii). The displayed relations also give the promised and the equality/strict-inequality variants stated after the theorem. The same threshold r_0^{rat} works for all a because it depends only on the fixed sampled order d_* , not on the unbounded denominators N_a . $\square$

6.3 Transcendental Turán densities

We now prove Theorem 1.2. The construction starts from a polynomial functional on probability-weighted rectangular 0–1 matrices. We compute its supremum as $\log 2 - 5/8$, approximate this value by finite Ferrers matrices, and then expand the functional in the 2×4 induced-density basis. An affine normalization makes it a valid compiler input.

Put

$$\mathfrak{B} := \mathfrak{B}_{2,4} = (\{\text{L}, \text{R}\}, (2, 4), \{\{\text{L}, \text{R}\}\}).$$

A finite $\mathfrak{B}$ -graph is a rectangular 0–1 matrix $M = (m_{pc})_{p \in \text{P}, c \in \text{C}}$. The sampling scheme draws two rows and four columns, although the finite row and column sets P and C may have arbitrary sizes. Equip them with probability weightings $x = (x_p)_{p \in \text{P}}$ and $y = (y_c)_{c \in \text{C}}$.

6.3.1 The matrix optimization

For $p, p' \in \mathbf{P}$, let

$$d_p := \sum_{c \in \mathbf{C}} y_c m_{pc}, \quad c_{pp'} := \sum_{c \in \mathbf{C}} y_c m_{pc} m_{p'c},$$

and set $u_{pp'} := d_p - c_{pp'}$, $v_{pp'} := d_{p'} - c_{pp'}$, and $w_{pp'} := c_{pp'}$. These are the y -masses of the column patterns 10, 01, and 11 for the ordered row pair (p, p') . In particular, they are nonnegative and have sum at most 1.

The functional below is designed to have two features. It penalizes incomparable row neighborhoods, while for nested neighborhoods its value depends only on the distribution of row degrees. That dependence becomes a separable quadratic functional of the stop-loss transform, whose pointwise optimizer produces log 2.

Define $V(s) := s^2/2 - s^3/6$. We use the full and truncated kernels

$$R_{\text{full}}(a, b) := \int_0^{\min(a, b)} (1+t)(a-t)(b-t) dt$$

and, for $0 \leq c \leq \min\{a, b\}$,

$$R(a, b, c) := \int_0^c (1+t)(a-t)(b-t) dt.$$

Direct integration gives

$$R(a, b, c) = abc + \frac{ab - a - b}{2} c^2 + \frac{1 - a - b}{3} c^3 + \frac{1}{4} c^4.$$

Thus $R(a, b, c)$ is the contribution of $[0, c]$ to the full kernel. Set

$$Q(u, v, w) := \frac{V(u+w) + V(v+w)}{2} - R(u+w, v+w, w) - uv$$

and define

$$J(M, x, y) := \sum_{p, p' \in \mathbf{P}} x_p x_{p'} Q(u_{pp'}, v_{pp'}, w_{pp'}).$$

The following kernel comparison shows that nonnested row neighborhoods can only decrease this functional.

Lemma 6.6. *For all $u, v, w \geq 0$ with $u + v + w \leq 1$,*

$$R(u+w, v+w, w) + uv \geq R_{\text{full}}(u+w, v+w).$$

Equality holds if and only if $uv = 0$.

Proof. By symmetry, assume $u \leq v$ and put $a := w + u$ and $b := w + v$. Since $\min\{a, b\} = a$,

$$R_{\text{full}}(a, b) - R(a, b, w) = \int_0^u (1+w+s)(u-s)(v-s) ds.$$

If $u = 0$, this difference and uv both vanish. If $u > 0$, then $v > 0$, $u < 1$, and $1 + w + s \leq 2$ for $0 \leq s \leq u$. Hence

$$R_{\text{full}}(a, b) - R(a, b, w) \leq 2 \int_0^u (u-s)(v-s) ds = u^2 v - \frac{u^3}{3} < uv.$$

This proves the inequality, and the strict final inequality shows that equality occurs exactly when $uv = 0$. $\square$

A rectangular 0–1 matrix is a *Ferrers matrix* if its row neighborhoods can be ordered by inclusion. For $z \in \mathbb{R}$, write $z_+ := \max\{z, 0\}$. The *stop-loss transform* of a $[0, 1]$ -valued random variable D is

$$H_D(t) := \mathbb{E}(D - t)_+ = \int_t^1 \Pr(D > s) ds, \quad 0 \leq t \leq 1,$$

where the second equality is the layer-cake identity.

For nested row neighborhoods, the common-neighborhood mass is the smaller row degree. The kernel comparison therefore reduces the matrix optimization to a one-variable variational problem for the stop-loss transform of the row-degree distribution.

Theorem 6.7. *One has*

$$\sup_{M, x, y} J(M, x, y) = \log 2 - \frac{5}{8},$$

where the supremum is over all finite rectangular 0–1 matrices and all probability weightings on their two sides. An explicit sequence of rationally weighted Ferrers matrices has J -values converging to this supremum.

Proof. Let I and I' be independent row indices with distribution x , and put $D := d_I$, $D' := d_{I'}$, and $H := H_D$. By Lemma 6.6,

$$J(M, x, y) \leq \mathbb{E} V(D) - \mathbb{E} R_{\text{full}}(D, D').$$

Since $V(0) = V'(0) = 0$ and $V''(t) = 1 - t$,

$$\mathbb{E} V(D) = \int_0^1 (1 - t) H(t) dt.$$

Fubini's theorem and the independence of D, D' similarly give

$$\mathbb{E} R_{\text{full}}(D, D') = \int_0^1 (1 + t) H(t)^2 dt.$$

Completing the square therefore yields

$$\begin{aligned} J(M, x, y) &\leq \int_0^1 ((1 - t)H(t) - (1 + t)H(t)^2) dt \\ &= \log 2 - \frac{5}{8} - \int_0^1 (1 + t) \left(H(t) - \frac{1 - t}{2(1 + t)} \right)^2 dt. \end{aligned}$$

Here the constant is obtained from

$$\frac{1}{4} \int_0^1 \frac{(1 - t)^2}{1 + t} dt = \log 2 - \frac{5}{8}.$$

It remains to attain the upper bound asymptotically. Put $H_*(t) := (1 - t)/(2(1 + t))$, and let D_∞ have density $2(1 + t)^{-3} dt$ on $[0, 1)$ together with an atom of mass $1/4$ at 1. The continuous part has mass $3/4$. Moreover, for $0 \leq s < 1$,

$$\Pr(D_\infty > s) = \int_s^1 \frac{2 du}{(1 + u)^3} + \frac{1}{4} = \frac{1}{(1 + s)^2}.$$

Consequently,

$$H_{D_\infty}(t) = \int_t^1 \frac{ds}{(1 + s)^2} = H_*(t).$$

We approximate this distribution by the row degrees of rationally weighted Ferrers matrices. For $m \geq 1$, let $M^{(m)}$ have rows indexed by $0 \leq k \leq m$, columns indexed by $1 \leq j \leq m$, and entries

$$m_{k,j}^{(m)} := \mathbf{1}_{\{j \leq k\}}.$$

Give the columns the uniform weighting $y^{(m)} = (1/m, \dots, 1/m)$ and assign to row k the weight

$$\xi_{m,k} := \begin{cases} \left(1 + \frac{k}{m}\right)^{-2} - \left(1 + \frac{k+1}{m}\right)^{-2}, & 0 \leq k < m, \\ \frac{1}{4}, & k = m. \end{cases}$$

Write $\xi_m := (\xi_{m,k})_{k=0}^m$. These weights are nonnegative rationals and telescope to 1. Row k has degree k/m , and the row neighborhoods are nested. Hence $M^{(m)}$ is Ferrers; for every ordered pair of rows, one of the 10- and 01-pattern masses vanishes. Thus equality holds in [Lemma 6.6](#) for every row pair.

Let D_m be the row degree under ξ_m . Couple it with D_∞ by rounding each value below 1 down to the nearest multiple of $1/m$ and leaving the atom at 1 fixed. Since

$$\Pr\left(\frac{k}{m} \leq D_\infty < \frac{k+1}{m}\right) = \xi_{m,k} \quad (0 \leq k < m),$$

the rounded variable has distribution D_m . Thus $|D_m - D_\infty| \leq 1/m$ and

$$\sup_{0 \leq t \leq 1} |H_{D_m}(t) - H_*(t)| \leq \frac{1}{m}.$$

For these Ferrers matrices, the preceding integral bound is an identity. Uniform convergence of the stop-loss transforms now gives

$$J(M^{(m)}, \xi_m, y^{(m)}) \longrightarrow \log 2 - \frac{5}{8},$$

which proves the reverse inequality. □

6.3.2 A normalized 2×4 input

We next express J in the 2×4 induced-density basis and normalize its coefficients into $(0, 1)$.

For $B \in \mathcal{U}_{\mathfrak{B}}$, choose a representative 2×4 matrix and order its two rows. Let $n_{00}, n_{10}, n_{01}, n_{11}$ count its four column types. Write

$$Q(u, v, w) = \sum_{i+j+k \leq 4} q_{ijk} u^i v^j w^k, \quad q_{ijk} \in \mathbb{Q},$$

and define

$$\sigma_B := \sum_{i+j+k \leq 4} q_{ijk} \frac{(n_{10})_i (n_{01})_j (n_{11})_k}{(4)_{i+j+k}}. \quad (62)$$

The required induced-density expansion follows from the factorial moments of the four sampled column types.

Lemma 6.8. *The rational number σ_B is effectively computable and is independent of both the choice of representative and the ordering of the rows. Let $q_J := \sum_{B \in \mathcal{U}_{\mathfrak{B}}} \sigma_B \mathbf{I}_{\mathfrak{B}}(B)$. For every finite weighted rectangular matrix,*

$$\widehat{t}_{\mathfrak{B}}(q_J; M, x, y) = J(M, x, y). \quad (63)$$

Proof. Permuting the columns does not change the four pattern counts. Interchanging the rows swaps n_{10} and n_{01} , but $Q(u, v, w) = Q(v, u, w)$ and hence $q_{ijk} = q_{jik}$. Thus (62) is independent of all choices. The same finite formula shows that σ_B is rational and effectively computable.

Let I_1, I_2 be independent row samples with distribution x , and let $C_1, \dots, C_4$ be independent column samples with distribution y . Conditional on $(I_1, I_2) = (p, p')$, the patterns 10, 01, and 11 have probabilities $u_{pp'}$, $v_{pp'}$, and $w_{pp'}$. If N_{10}, N_{01}, N_{11} are their respective counts, then

$$\mathbb{E}[(N_{10})_i (N_{01})_j (N_{11})_k \mid I_1 = p \text{ and } I_2 = p'] = (4)_{i+j+k} u_{pp'}^i v_{pp'}^j w_{pp'}^k.$$

Substitution into (62), followed by averaging over the ordered row pair, gives (63). $\square$

For the normalization, put $\mathbf{1}_{\mathfrak{B}} := \sum_{B \in \mathcal{U}_{\mathfrak{B}}} \mathbf{I}_{\mathfrak{B}}(B)$, whose evaluation is identically 1, and choose

$$C_{\text{tr}} := 1 + \max_{B \in \mathcal{U}_{\mathfrak{B}}} [|\sigma_B|].$$

Define

$$q_{\text{tr}} := \frac{1}{2} \mathbf{1}_{\mathfrak{B}} + \frac{1}{2C_{\text{tr}}} q_J = \sum_{B \in \mathcal{U}_{\mathfrak{B}}} \left(\frac{1}{2} + \frac{\sigma_B}{2C_{\text{tr}}} \right) \mathbf{I}_{\mathfrak{B}}(B). \quad (64)$$

Every coefficient lies strictly between 0 and 1. Let N_{tr} be the least common multiple of their denominators; it is computable from this finite list. By Lemma 6.8 and Theorem 6.7,

$$\text{Val}_{\mathfrak{B}}(q_{\text{tr}}) = \frac{1}{2} + \frac{1}{2C_{\text{tr}}} \left(\log 2 - \frac{5}{8} \right) = \alpha_{\text{tr}} + \beta_{\text{tr}} \log 2, \quad (65)$$

where

$$\alpha_{\text{tr}} := \frac{1}{2} - \frac{5}{16C_{\text{tr}}}, \quad \beta_{\text{tr}} := \frac{1}{2C_{\text{tr}}}.$$

Both are rational, and $\beta_{\text{tr}} > 0$.

Proof of Theorem 1.2. Set $r_0^{\text{tr}} := \max\{3, r_{\text{aff}}(6)\}$. For $r \geq r_0^{\text{tr}}$, apply Theorem 1.3 to the scheme $\mathfrak{B}_{2,4}$, denominator N_{tr} , and input q_{tr} , and denote the resulting family by $\mathcal{F}_r$. Set

$$A_r := A_{\mathfrak{B}_{2,4}, r, N_{\text{tr}}} + B_{\mathfrak{B}_{2,4}, r, N_{\text{tr}}} \alpha_{\text{tr}}, \quad B_r := B_{\mathfrak{B}_{2,4}, r, N_{\text{tr}}} \beta_{\text{tr}}.$$

Then $A_r \in \mathbb{Q}$, $B_r \in \mathbb{Q}_{>0}$, and (65) gives $\pi(\mathcal{F}_r) = A_r + B_r \log 2$. Since $\log 2$ is transcendental by the Hermite–Lindemann theorem [3, Chapter 1, §3], the same holds for $\pi(\mathcal{F}_r)$. $\square$

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Declaration on the use of AI

The authors used generative AI tools to discuss the organization of the proofs, check proof details, and improve the exposition.

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